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Nezami et al.

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(54) **TRAINING DATA AUGMENTATION USING GAZETTEERS AND PERTURBATIONS TO FACILITATE TRAINING NAMED ENTITY RECOGNITION MODELS**

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G06F 40/295	(2020.01)
G06N 3/006	(2023.01)

(52) **U.S. Cl.**
CPC ***G06F 40/295*** (2020.01); ***G06N 3/006*** (2013.01)

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(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2022/0222489 A1* 7/2022 Liu G06F 18/214

FOREIGN PATENT DOCUMENTS

CN 111554278 A * 8/2020

OTHER PUBLICATIONS

Lara-Clares et al., LSI2 Uned at eHealth-KD Challenge 2019 a Few-shot Learning Model for Knowledge Discovery from eHealth Documents, CEUR Workshop Proceedings, Sep. 24, 2019, pp. 60-66.

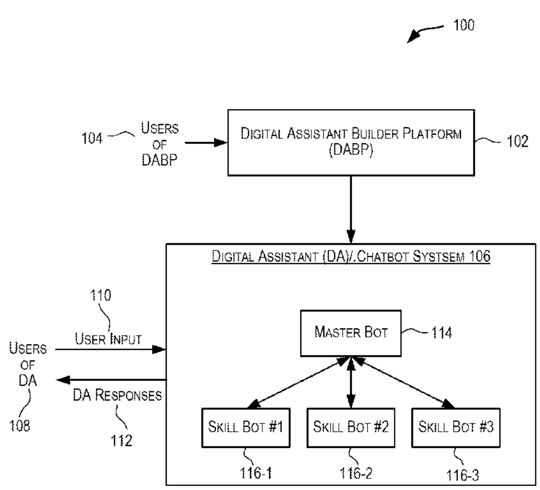
(Continued)

Primary Examiner — Md S Elahee
(74) *Attorney, Agent, or Firm* — Kilpatrick Townsend & Stockton LLP

(57) **ABSTRACT**

Techniques are provided for augmenting training data using gazetteers and perturbations to facilitate training named entity recognition models. The training data can be augmented by generating additional utterances from original utterances in the training data and combining the generated additional utterances with the original utterances to form the augmented training data. The additional utterances can be generated by replacing the named entities in the original utterances with different named entities and/or perturbed versions of the named entities in the original utterances selected from a gazetteer. Gazetteers of named entities can be generated from the training data and expanded by searching a knowledge base and/or perturbing the named entities therein. The named entity recognition model can be trained using the augmented training data.

20 Claims, 10 Drawing Sheets



(58) **Field of Classification Search**
USPC 704/9
See application file for complete search history.

(56) **References Cited**

OTHER PUBLICATIONS

Peshterliev et al., Self-Attention Gazetteer Embeddings for Named-Entity Recognition, Computer Science, Available Online at: https://arxiv.org/pdf/2004.04060.pdf, Apr. 18, 2020, 6 pages.
Runge et al., Exploring Neural Entity Representations for Semantic Information, Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP, Nov. 17, 2020, 13 pages.
Sang et al., Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition, Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003, Jun. 12, 2003, 6 pages.
Song et al., Gazetteer Generation for Neural Named Entity Recognition, The Thirty-Third International FLAIRS Conference (FLAIRS-33), May 2020, pp. 298-301.
Yamada et al., Wikipedia2Vec: An Efficient Toolkit for Learning and Visualizing the Embeddings of Words and Entities from Wikipedia, Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, Sep. 26, 2020, 8 pages.

* cited by examiner

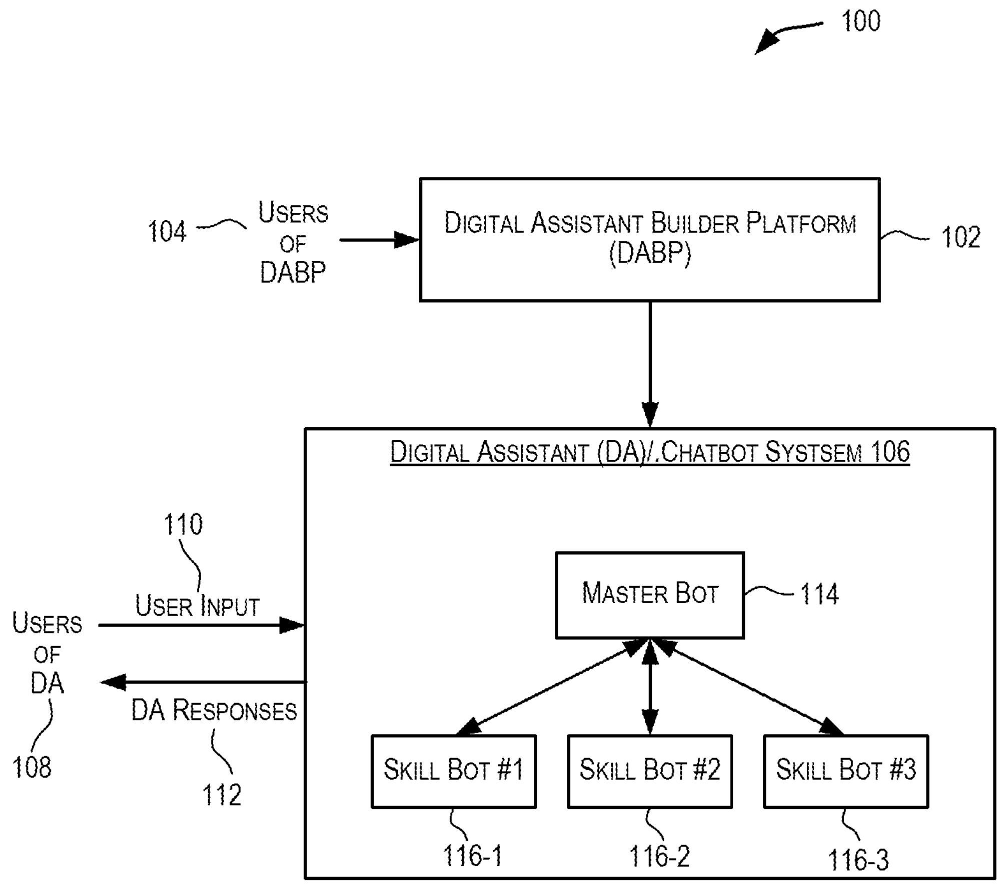

100
104
USERS OF DABP
DIGITAL ASSISTANT BUILDER PLATFORM (DABP)
102
DIGITAL ASSISTANT (DA)/.CHATBOT SYSTSEM 106
110
USER INPUT
USERS OF DA
DA RESPONSES
108
112
MASTER BOT
114
SKILL BOT #1
SKILL BOT #2
SKILL BOT #3
116-1
116-2
116-3

FIG. 1

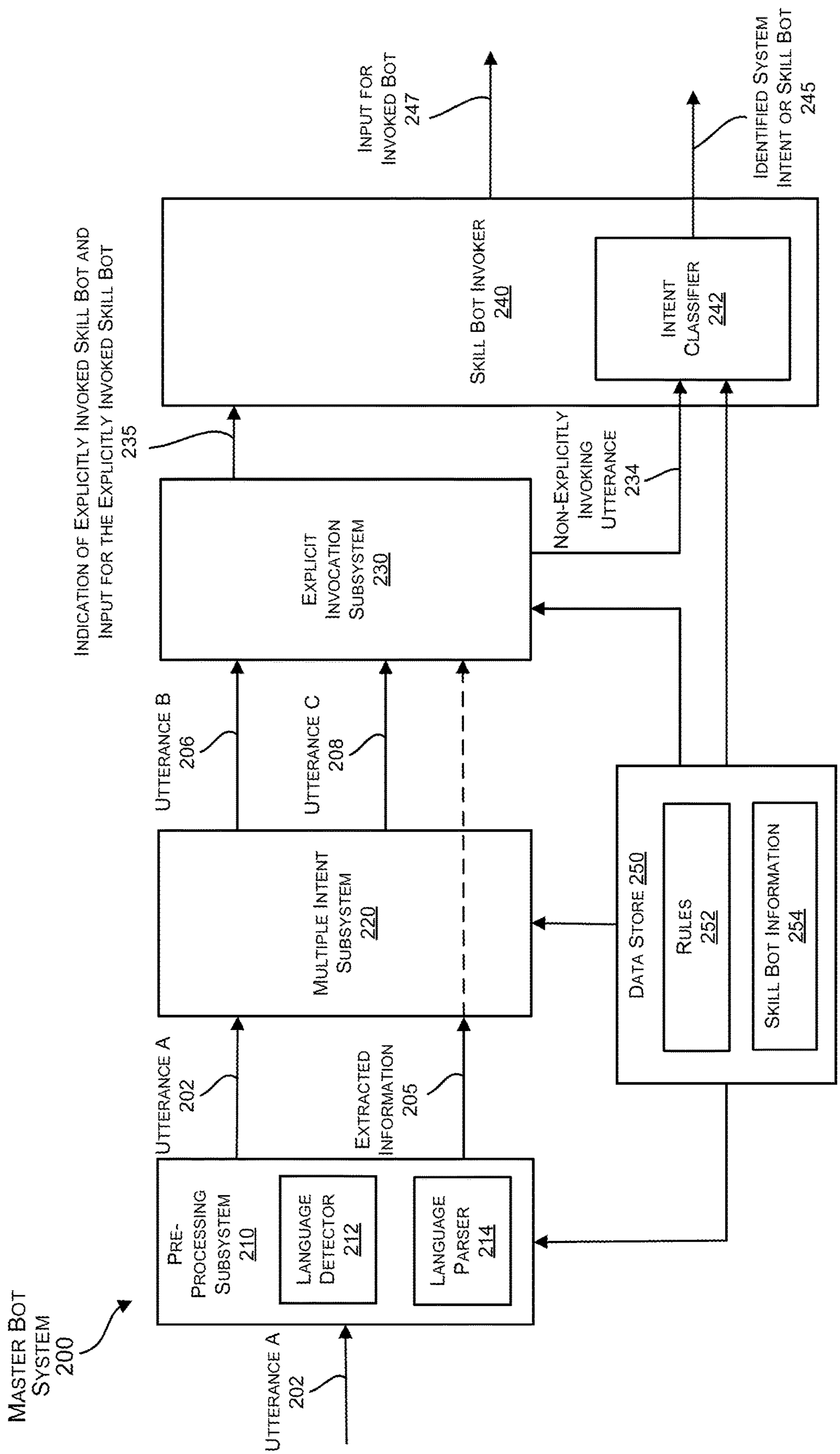
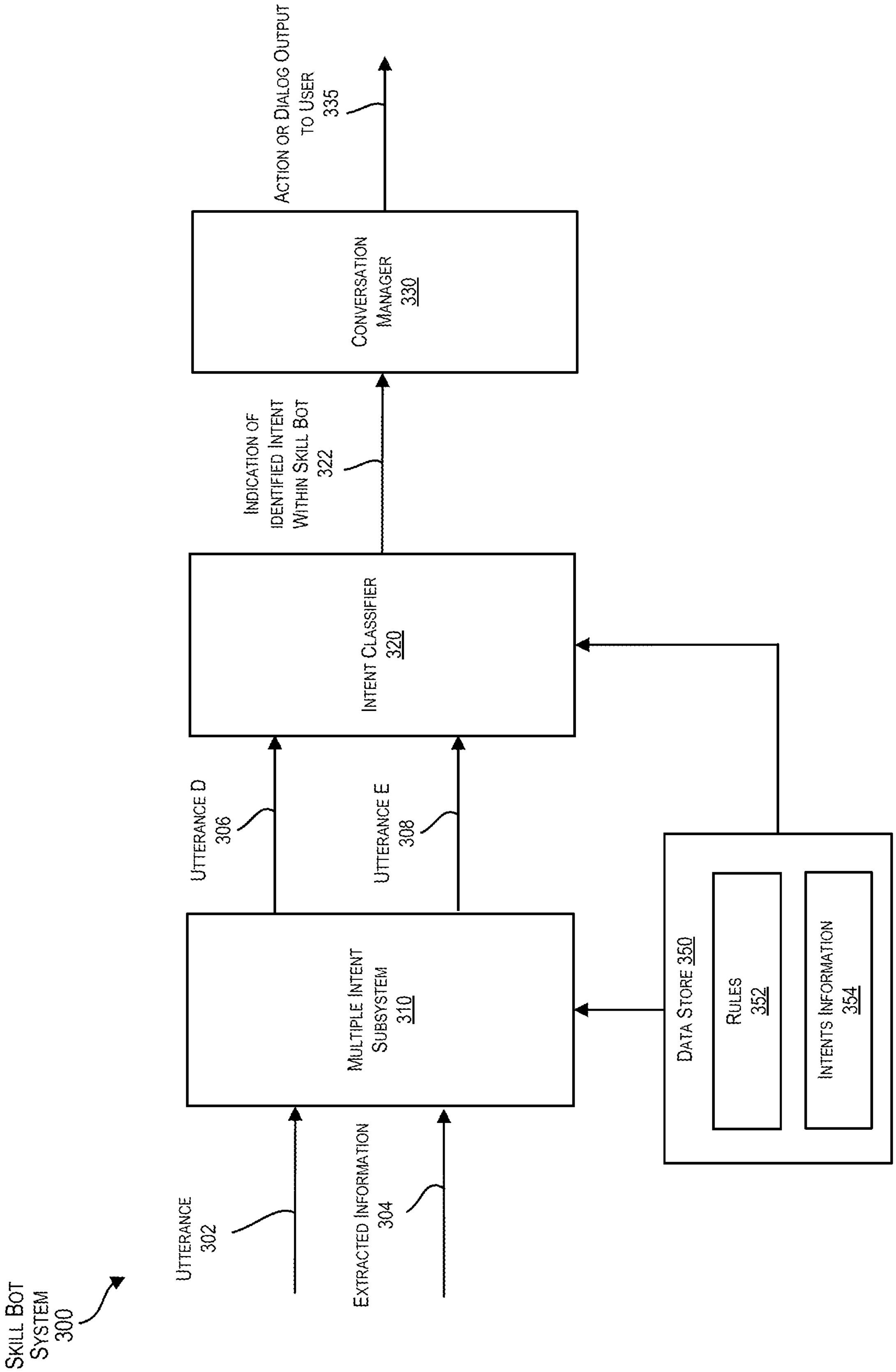
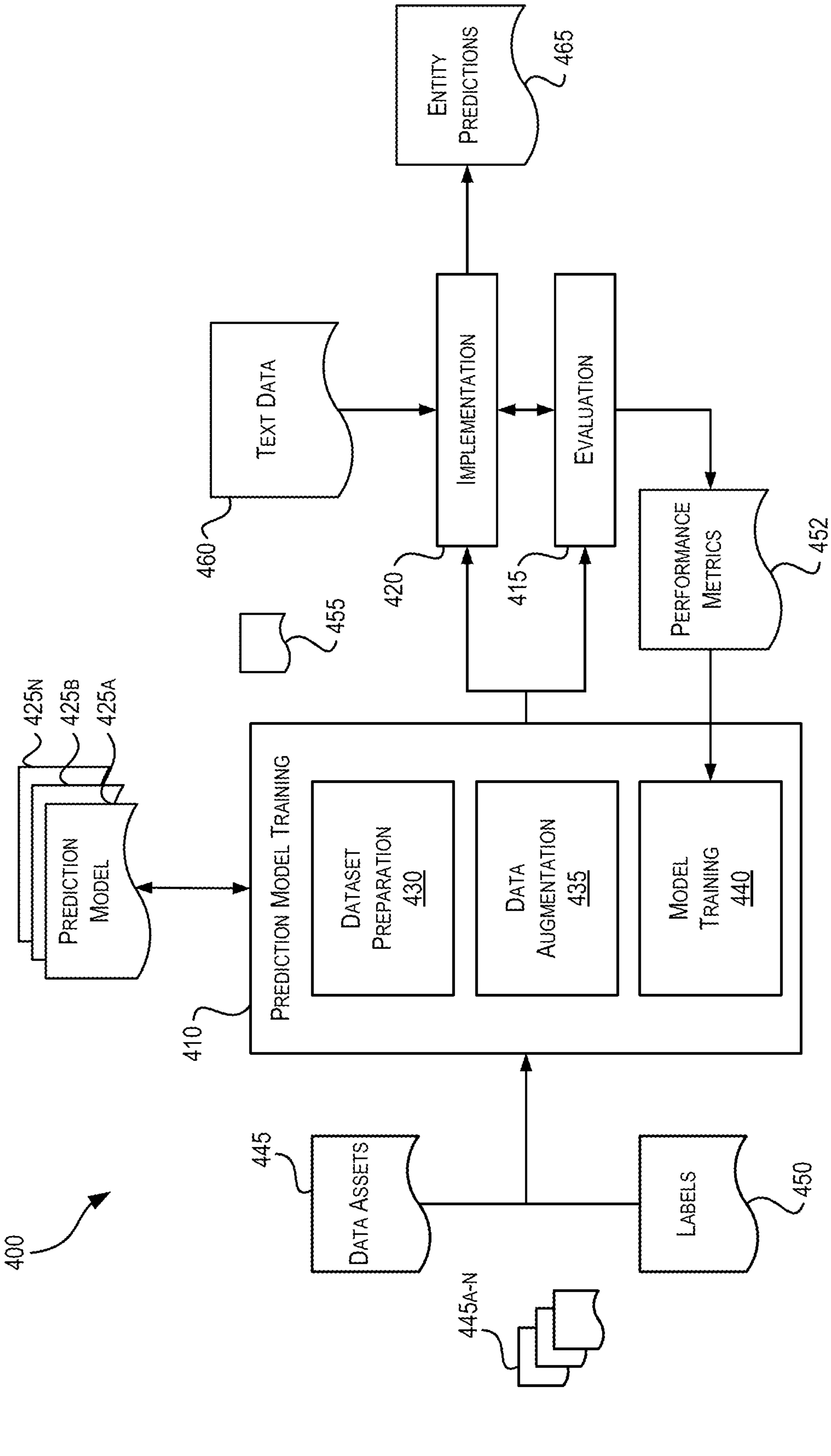

Master Bot System 200
Utterance A 202
Pre-Processing Subsystem 210
Language Detector 212
Language Parser 214
Utterance A 202
Extracted Information 205
Multiple Intent Subsystem 220
Utterance B 206
Utterance C 208
Explicit Invocation Subsystem 230
Indication of Explicitly Invoked Skill Bot and Input for the Explicitly Invoked Skill Bot 235
Non-Explicitly Invoking Utterance 234
Skill Bot Invoker 240
Intent Classifier 242
Input for Invoked Bot 247
Identified System Intent or Skill Bot 245
Data Store 250
Rules 252
Skill Bot Information 254

FIG. 2



Skill Bot System 300
Utterance 302
Extracted Information 304
Multiple Intent Subsystem 310
Utterance D 306
Utterance E 308
Intent Classifier 320
Indication of Identified Intent Within Skill Bot 322
Conversation Manager 330
Action or Dialog Output to User 335
Data Store 350
Rules 352
Intents Information 354

FIG. 3



400
Data Assets
445
445A-N
Labels
450
Prediction Model Training
410
Dataset Preparation 430
Data Augmentation 435
Model Training 440
Prediction Model
425N
425B
425A
455
Text Data
460
Implementation
420
Evaluation
415
Performance Metrics
452
Entity Predictions
465

FIG. 4A

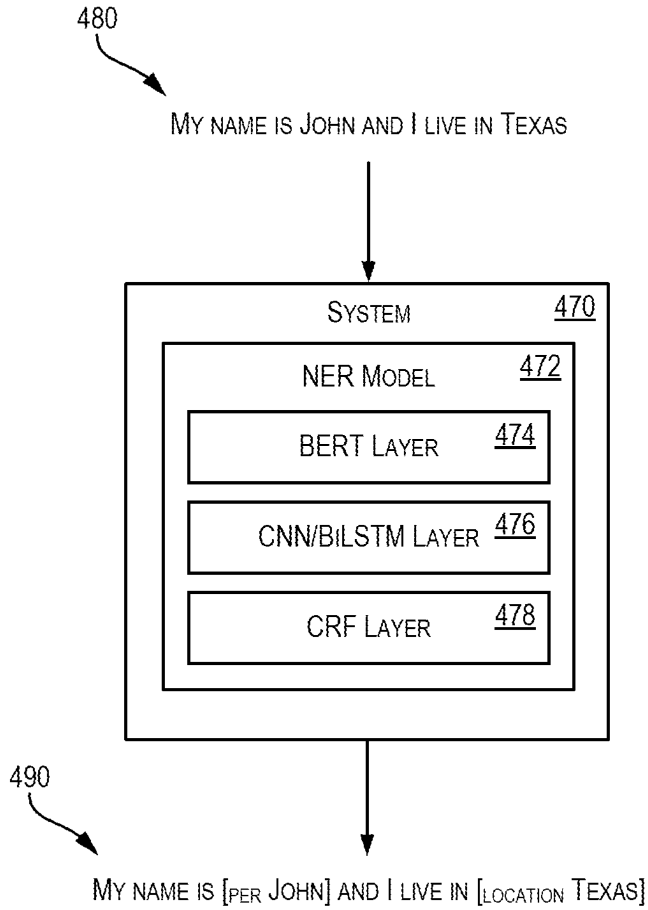

480
MY NAME IS JOHN AND I LIVE IN TEXAS
SYSTEM
470
NER MODEL
472
BERT LAYER
474
CNN/BILSTM LAYER
476
CRF LAYER
478
490
MY NAME IS [PER JOHN] AND I LIVE IN [LOCATION TEXAS]

FIG. 4B

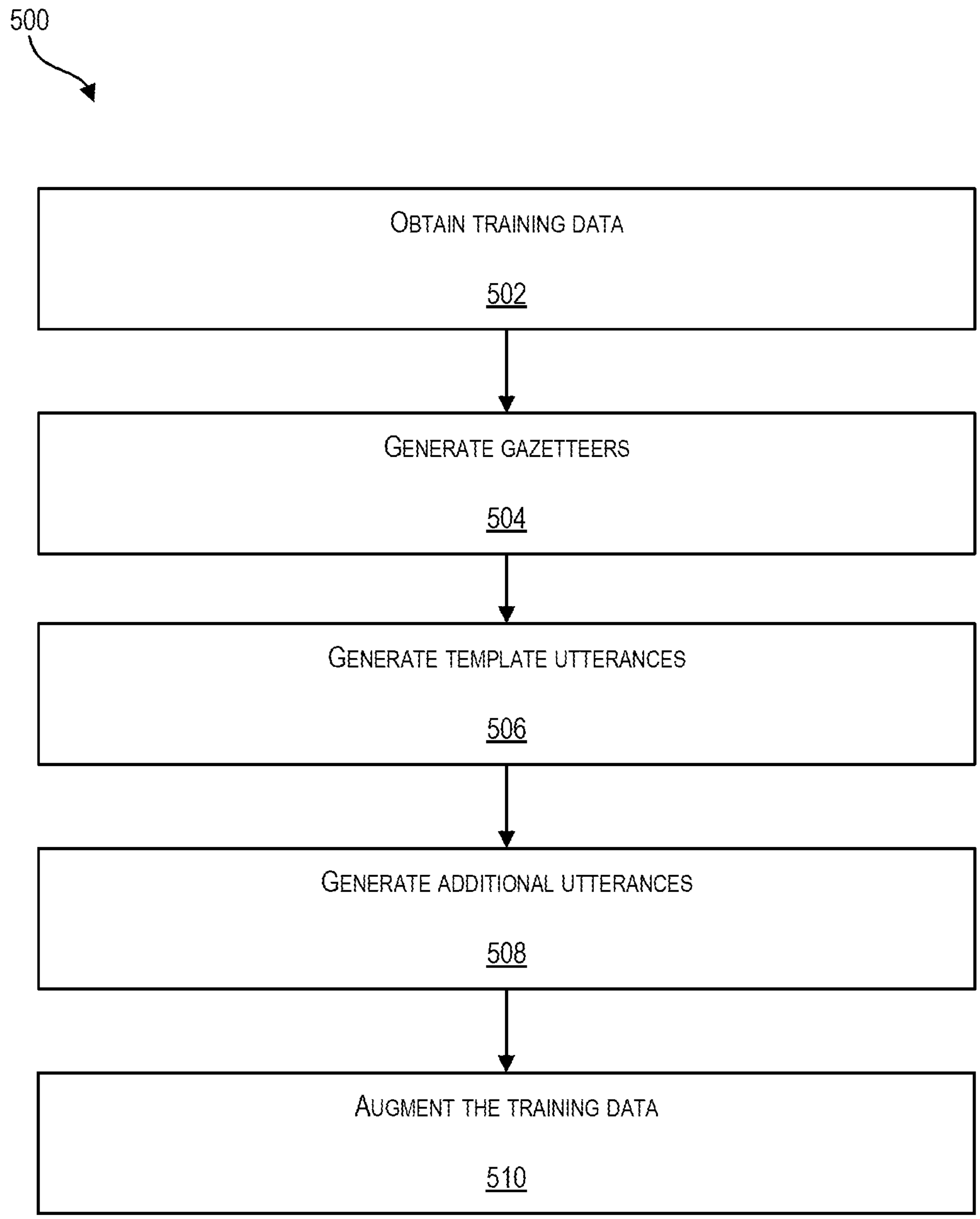

500
OBTAIN TRAINING DATA
502
GENERATE GAZETTEERS
504
GENERATE TEMPLATE UTTERANCES
506
GENERATE ADDITIONAL UTTERANCES
508
AUGMENT THE TRAINING DATA
510

FIG. 5A

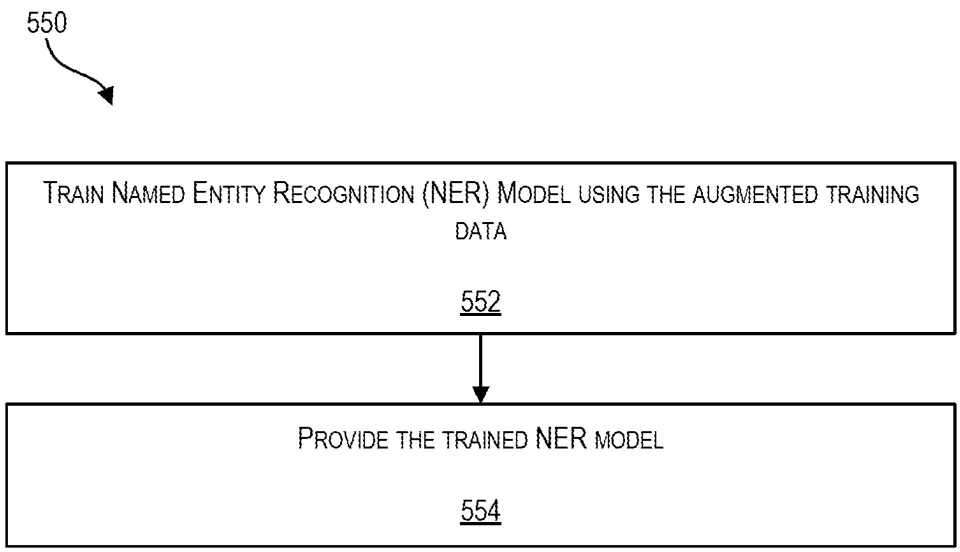

550
TRAIN NAMED ENTITY RECOGNITION (NER) MODEL USING THE AUGMENTED TRAINING DATA
552
PROVIDE THE TRAINED NER MODEL
554

FIG. 5B

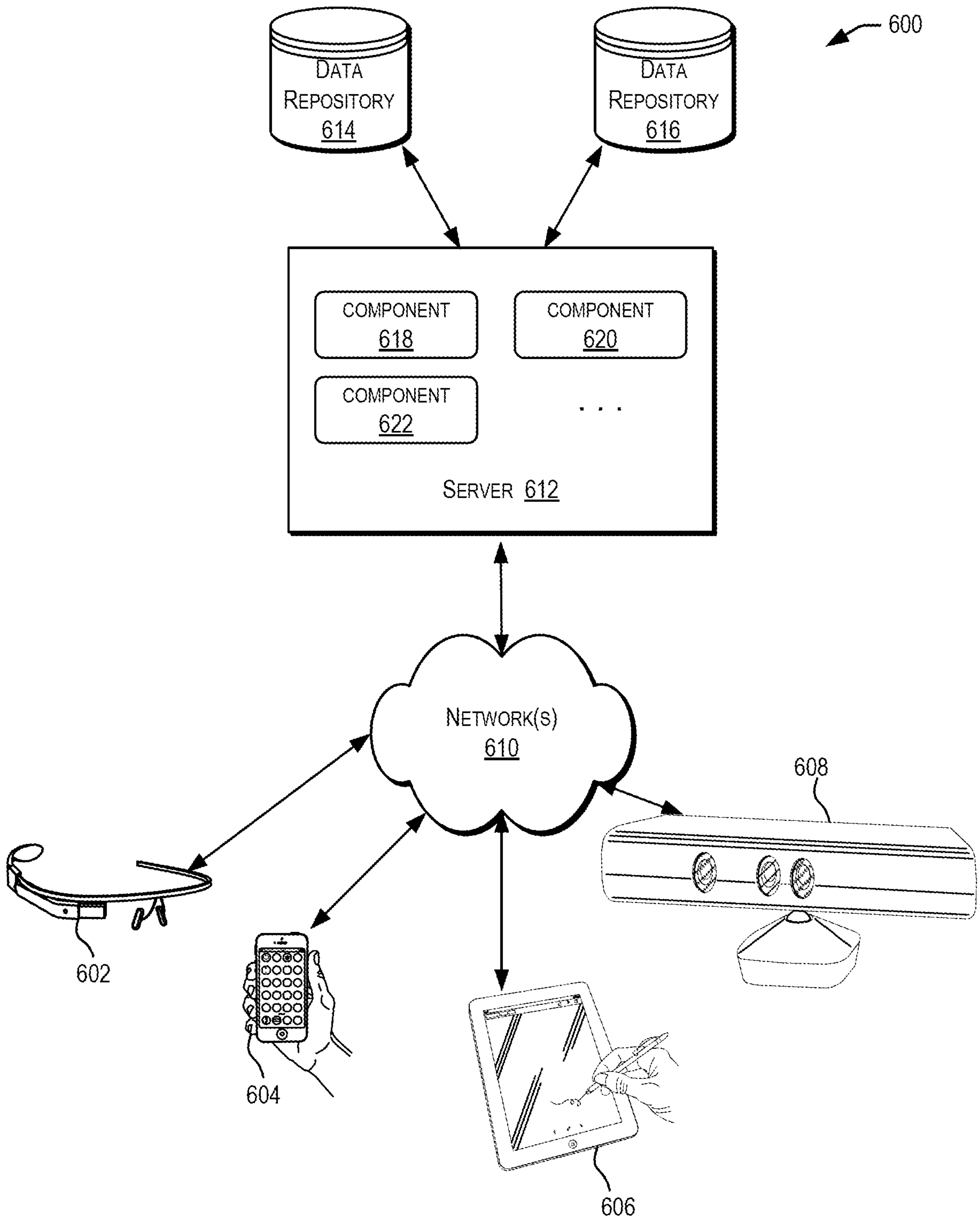

600
DATA REPOSITORY 614
DATA REPOSITORY 616
COMPONENT 618
COMPONENT 620
COMPONENT 622
. . .
SERVER 612
NETWORK(S) 610
602
604
606
608

FIG. 6

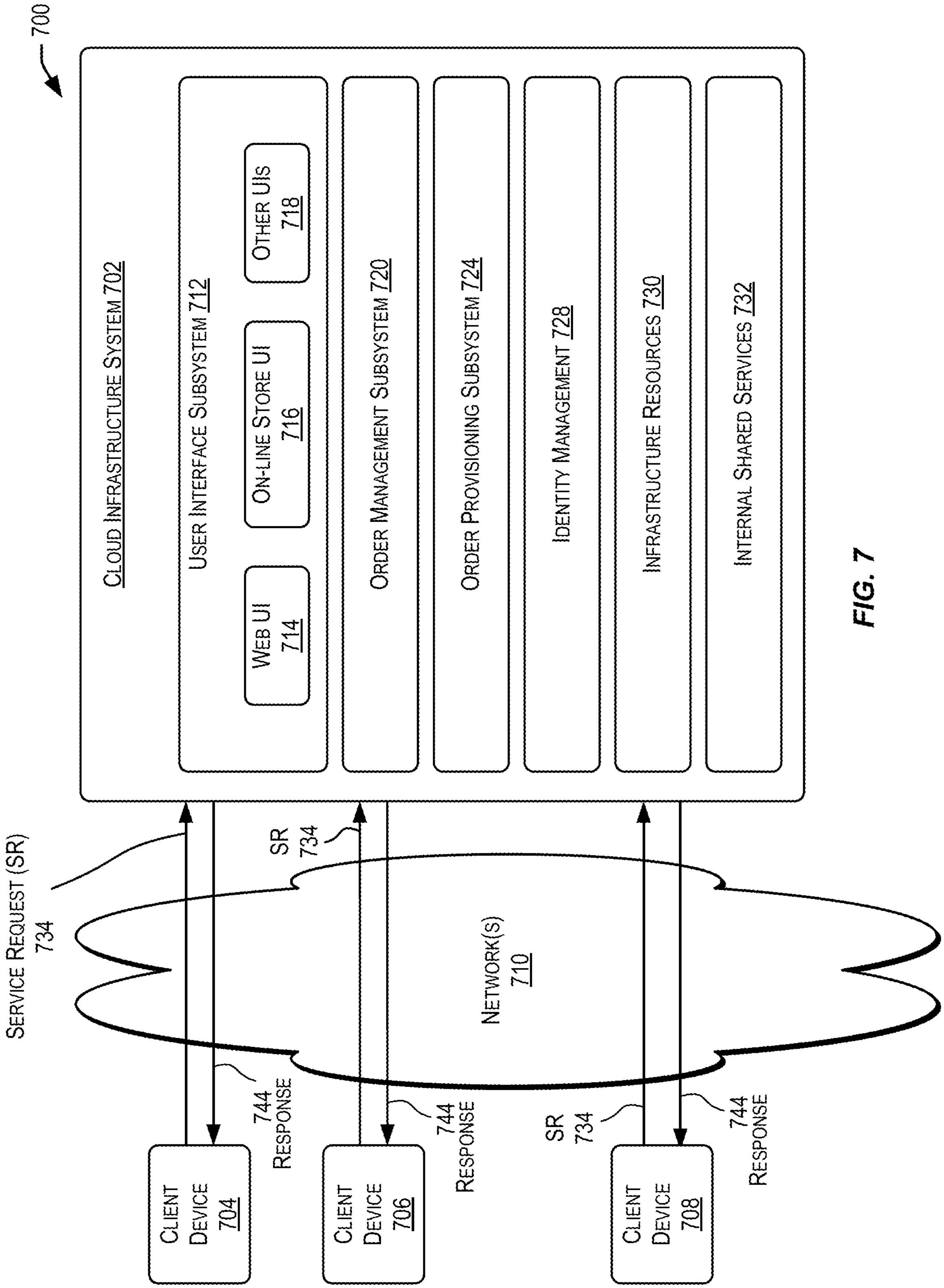

700
Cloud Infrastructure System 702
User Interface Subsystem 712
Web UI 714
On-line Store UI 716
Other UIs 718
Order Management Subsystem 720
Order Provisioning Subsystem 724
Identity Management 728
Infrastructure Resources 730
Internal Shared Services 732
Service Request (SR) 734
SR 734
SR 734
Network(s) 710
744 Response
744 Response
744 Response
Client Device 704
Client Device 706
Client Device 708

FIG. 7

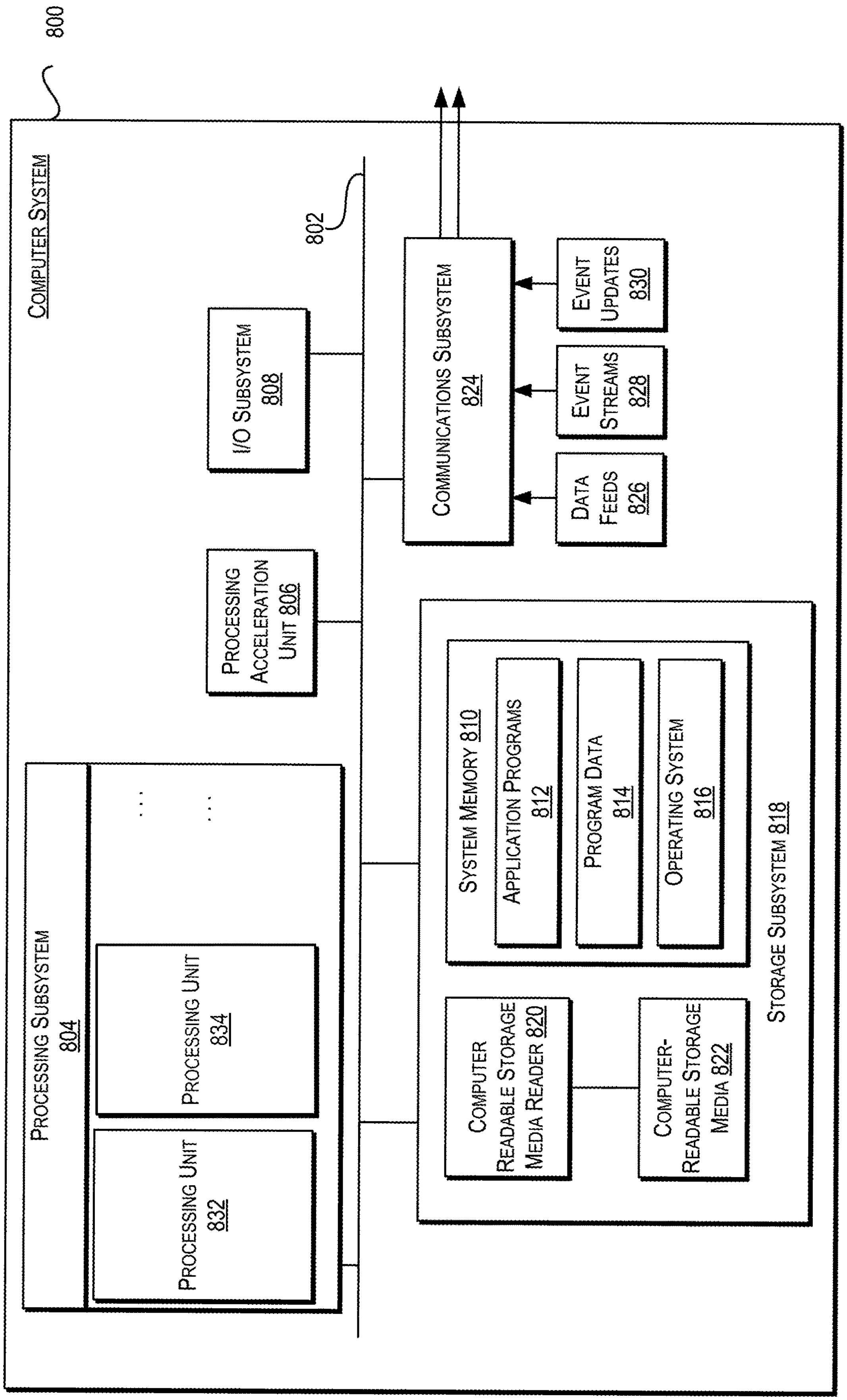

800
Computer System
802
I/O Subsystem 808
Processing Acceleration Unit 806
Processing Subsystem 804
Processing Unit 832
Processing Unit 834
Communications Subsystem 824
Data Feeds 826
Event Streams 828
Event Updates 830
System Memory 810
Application Programs 812
Program Data 814
Operating System 816
Storage Subsystem 818
Computer Readable Storage Media Reader 820
Computer-Readable Storage Media 822

FIG. 8

TRAINING DATA AUGMENTATION USING GAZETTEERS AND PERTURBATIONS TO FACILITATE TRAINING NAMED ENTITY RECOGNITION MODELS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a non-provisional application of and claims the benefit of and priority to under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 63/362,233 having a filing date of Mar. 31, 2022, and U.S. Provisional Application No. 63/362,234 having a filing date of Mar. 31, 2022, the entire contents of which are incorporated herein by reference for all purposes.

FIELD

The present disclosure relates generally to artificial intelligence techniques, and more particularly, to techniques for augmenting training data using gazetteers and perturbations to facilitate training named entity recognition models.

BACKGROUND

Artificial intelligence has many applications. To illustrate, many users around the world use instant messaging or chat platforms in order to get instant reactions. Organizations often use these instant messaging or chat platforms to engage with customers (or end users) in live conversations. However, it can be very costly for organizations to employ service people to engage in live communication with customers or end users. Chatbots or bots have been developed to simulate conversations with end users, especially over the Internet. End users can communicate with such bots through messaging apps. An intelligent bot, generally a bot powered by artificial intelligence (AI), can communicate intelligently and contextually in live conversations with end users, which allows for a more natural conversation and an improved conversational experience. Instead of relying on a fixed set of keywords or commands, intelligent bots may be able to receive utterances of end users in natural language, understand their intentions, and respond accordingly.

However, artificial intelligence-based solutions, such as chatbots, can be difficult to build because these automated solutions require specific knowledge in certain fields and the application of certain techniques that may be solely within the capabilities of specialized developers. As part of building such chatbots, a developer may first understand the needs of enterprises and end users. The developer may then analyze and make decisions related to, for example, selecting data sets to be used for the analysis, preparing the input data sets for analysis (e.g., cleansing the data, extracting, formatting, and/or transforming the data prior to analysis, performing data features engineering, etc.), identifying an appropriate machine learning (ML) technique(s) or model(s) for performing the analysis, and improving the technique or model to improve results/outcomes based upon feedback. The task of identifying an appropriate model may include developing multiple models, possibly in parallel, iteratively testing and experimenting with these models, before identifying a particular model (or models) for use. Further, supervised learning-based solutions typically involve a training phase, followed by an application (i.e., inference) phase, and iterative loops between the training phase and the application phase. The developer may be responsible for carefully implementing and monitoring these phases to achieve optimal solutions. For example, to train the ML technique(s) or model(s), precise training data is required to enable the algorithms to understand and learn certain patterns or features (e.g., for chatbots-intent extraction and careful syntactic analysis, not just raw language processing) that the ML technique(s) or model(s) will use to predict the outcome desired (e.g., inference of an intent from an utterance). In order to ensure the ML technique(s) or model(s) learn these patterns and features properly, the developer may be responsible for selecting, enriching, and optimizing sets of training data for the ML technique(s) or model(s).

BRIEF SUMMARY

Techniques are disclosed herein for augmenting training data using gazetteers and perturbations to facilitate training named entity recognition (NER) models.

In various embodiments, a computer-implemented method includes accessing training data comprising a plurality of original utterances, wherein each original utterance of the plurality of original utterances comprises at least one named entity corresponding to a named entity category of a plurality of named entity categories; accessing one or more gazetteers, wherein each gazetteer of the one or more gazetteers comprises a plurality of named entities extracted from the plurality of original utterances and a plurality of perturbed named entities derived from one or more named entities of the plurality of named entities; generating a plurality of template utterances, wherein each template utterance of the plurality of template utterances comprises information from an original utterance of the plurality of original utterances and at least one placeholder identifier representing a named entity in the original utterance of the plurality of original utterances; generating a plurality of additional utterances, wherein each additional utterance of the plurality of additional utterances comprises a template utterance of the plurality of template utterances populated with at least one named entity selected from a gazetteer of the one or more gazetteers; augmenting the training data by adding the plurality of additional utterances to the plurality of original utterances; and training a NER model with the augmented training data.

In some embodiments, each gazetteer of the one or more gazetteers corresponds to a different named entity category of the plurality of named entity categories.

In some embodiments, at least one gazetteer of the one or more gazetteers comprises a plurality of named entities retrieved from a source other than the training data.

In some embodiments, the plurality of named entities retrieved from the source other than the training data comprises named entities retrieved using at least one of a pre-trained model and a query-based search.

In some embodiments, the source other than the training data comprises at least one knowledge base.

In some embodiments, the plurality of perturbed named entities derived from one or more named entities of the plurality of named entities comprises perturbed versions of named entities of the plurality of named entities.

In some embodiments, a particular perturbed version of the perturbed versions of named entities of the plurality of named entities comprises a named entity having at least one typographical error.

In some embodiments, the computer-implemented method further includes providing the trained NER model to a system, wherein the providing the trained NER model includes detecting and classifying named entities in utterances received by the system from a user.

Some embodiments include a system including one or more processors and one or more computer-readable media storing instructions which, when executed by the one or more processors, cause the system to perform part or all of the operations and/or methods disclosed herein.

Some embodiments include one or more non-transitory computer-readable media storing instructions which, when executed by one or more processors, cause a system to perform part or all of the operations and/or methods disclosed herein.

The techniques described above and below may be implemented in a number of ways and in a number of contexts. Several example implementations and contexts are provided with reference to the following figures, as described below in more detail. However, the following implementations and contexts are but a few of many.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a simplified block diagram of a distributed environment incorporating an exemplary embodiment.

FIG. **2** is a simplified block diagram of a computing system implementing a master bot according to certain embodiments.

FIG. **3** is a simplified block diagram of a computing system implementing a skill bot according to certain embodiments.

FIG. **4**A is a simplified block diagram of a model training and deployment system according to certain embodiments.

FIG. **4**B is a simplified block diagram of a system incorporating a named entity recognition model according to certain embodiments.

FIG. **5**A is a flowchart that illustrates an example process for augmenting training data according to certain embodiments.

FIG. **5**B is a flowchart that illustrates an example process for providing a named entity recognition model according to certain embodiments.

FIG. **6** depicts a simplified diagram of a distributed system for implementing various embodiments.

FIG. **7** is a simplified block diagram of one or more components of a system environment by which services provided by one or more components of an embodiment system may be offered as cloud services, in accordance with various embodiments.

FIG. **8** illustrates an example computer system that may be used to implement various embodiments.

DETAILED DESCRIPTION

In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of certain embodiments. However, it will be apparent that various embodiments may be practiced without these specific details. The figures and description are not intended to be restrictive. The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any embodiment or design described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments or designs.

Introduction

Artificial intelligence techniques have broad applicability. For example, a digital assistant is an artificially intelligent driven interface that helps users accomplish a variety of tasks in natural language conversations. For each digital assistant, a customer may assemble one or more skills. Skills (also described herein as chatbots, bots, or skill bots) are individual bots that are focused on specific types of tasks, such as tracking inventory, submitting timecards, and creating expense reports. When an end user engages with the digital assistant, the digital assistant evaluates the end user input and routes the conversation to and from the appropriate chatbot. The digital assistant can be made available to end users through a variety of channels such as FACEBOOK® Messenger, SKYPE MOBILE® messenger, or a Short Message Service (SMS). Channels carry the chat back and forth from end users on various messaging platforms to the digital assistant and its various bots. The channels may also support user agent escalation, event-initiated conversations, and testing.

Intents allow artificial intelligence-based technology such as a chatbot to understand what the user wants the chatbot to do. Intents refer to the user's intention communicated to the chatbot via user requests and statements, which are also referred to as utterances (e.g., get account balance, make a purchase, etc.). As used herein, an utterance or a message may refer to a set of words (e.g., one or more sentences) exchanged during a conversation with a chatbot. Intents may be created by providing a name that illustrates some user action (e.g., order a pizza) and compiling a set of real-life user statements, or utterances that are commonly associated with triggering the action. Because the chatbot's cognition is derived from these intents, each intent may be created from a data set that is robust (one to two dozen utterances) and varied, so that the chatbot may interpret ambiguous user input. A rich set of utterances enables a chatbot to understand what the user wants when it receives messages like "Forget this order!" or "Cancel delivery!"—messages that mean the same thing but are expressed differently. Intent classifiers are included in chatbot systems to automatically classify intents of the utterances.

Utterances may include named entities. In addition to intention, named entities further allow a chatbot to understand the meaning of the utterances because the named entities modify the intent(s). For example, if a user types "show me yesterday's financial news," the named entities "yesterday" and "financial" assist the chatbot in understanding the user's request. Entities may be categorized according to what they represent. For example, "yesterday" may be categorized as "dateTime" and "financial" may be categorized as "newsType." Entities are sometimes referred to as slots. Named entity recognition (NER) is a tool used by the intent classifiers and chatbot systems to automatically detect, extract, and classify entities. Collectively, the utterances, including the named entities and intents that belong to them, make up a training corpus for the chatbot. By training an algorithm with the corpus, a customer turns that algorithm into a model that serves as a reference tool for resolving end user input(s) to a single intent. A customer can improve the acuity of the chatbot's cognition through rounds of intent testing and intent training.

Utilization of artificial intelligence in the context of chatbots illustrates some of the challenges of the application of artificial intelligence techniques. For example, building a chatbot that can determine end users' intents based upon the end users' utterances is a challenging task at least due to the subtleties and ambiguity of natural languages and the dimensions of the input/output space (e.g., possible user utterances, number of intents, etc.). An illustrative example of this difficulty arises from characteristics of natural language, such as employing euphemisms, synonyms, or ungrammatical speech to express intent. For example, an utterance may express an intent to order a pizza without explicitly mentioning the words pizza, ordering, or delivery. These characteristics of natural language give rise to uncertainty and result in chatbots using confidence as a parameter for prediction of user intents. As such, chatbots may need to be trained, monitored, debugged, and retrained in order to improve the performance of the chatbot and user experience with the chatbot. In conventional spoken language understanding (SLU) and natural language processing (NLP) systems, training mechanisms are provided for training and retraining machine-learning algorithms of the digital assistant or chatbot included therein. Conventionally, these algorithms are trained with "manufactured" utterances for any intent. For example, the utterance "Do you do price changes?" may be used to train a classification algorithm of a chatbot system to classify this type of utterance as the intent—"Do you offer a price match." The training of algorithms with manufactured utterances helps initially train the chatbot system for providing services and re-train the chatbot system once it is deployed and receives utterances from users.

As discussed above, artificial intelligence-based technology such as a chatbot can be trained to understand the meaning of utterances, which involves identifying one or more intents and one or more entities for the respective utterances. Entity extraction generally has two phases: a named entity recognition phase and an entity resolution phase. The particular problem addressed here pertains to the named entity recognition phase. In some cases, entities are typically things like date, time, locations, names, brands etc. In other cases, entities are system entities in that they are common and domain independent (e.g., PERSON, NUMBER, CURRENCY, DATE_TIME). In the named entity recognition phase, one or more models for named entity recognition are typically used detect and classify named entities in the respective utterances. An entity is often detected and classified as a named entity because of contextual information in the respective utterances (i.e., information in the respective utterances other than the named entities). In some cases, an entity name is context dependent. Conventional training of NER models starts with pre-labeled data. In a supervised machine-learning setting, particularly in this problem, the challenge is centered around the lack of sufficient training data with pre-labeled data for the NER models to learn from. As a result, conventional NER models tend to pay more attention to entities in the respective utterances than to the contextual information of the entity, which results in poor performance on utterances associated with entities which are not in the training data (i.e., conventional NER models poorly generalize to new and unseen entities). Any machine-learning model is only as good as the training data it was trained on. Thus, the training data quality is determinative of the model behavior.

Conventional NER models are typically trained using publicly and privately available datasets. However, these datasets are often not diverse enough to train the NER models to detect entities with all kinds of context and entity variations. One option for diversifying these datasets is to write additional labeled utterances oneself and add them to the datasets. Another option is to outsource the writing of additional utterances to freelancers and/or companies dedicated to data labeling and the like. Another option is to use crowd sourcing, which essentially scales the manual work using crowd workers. These approaches however can be difficult to implement for enterprise systems that employ many chatbot systems trained for many different tasks in multiple languages and are receiving a wide variety of utterances for each task. In systems employing chatbots such as these, the diversity in entities needs to be obtained automatically in a synthetic agnostic manner to quickly and efficiently generate large corpora of training data in multiple languages for many different chatbots.

Accordingly, a different approach is needed to address these challenges and others. The developed approach utilizes gazetteers and perturbations to augment the training data. The training data includes utterances with each utterance including at least one named entity corresponding to a named entity category and contextual information. The developed approach augments the training data by generating additional utterances from original utterances in the training data and combining the generated additional utterances with the original utterances to form the augmented training data. A NER model can be trained with the augmented training data, which results in improved performance of the NER model. Using the training data augmented as described herein, a NER model can learn to consider context in detecting and classifying named entities in utterances even when the named entities in utterances do not match the named entities in the training data. Additionally, the NER model can learn to detect and classify named entities in the utterances even if the named entities in the utterances include typographical errors. In this way, a NER model trained using the training data augmented with the techniques described herein can generalize well to variations of named entities and new and unseen named entities.

At a high level, the developed approach generates the additional utterances from the original utterances in the training data by maintaining the contextual information of the original utterances and replacing the named entities in the original utterances with different named entities selected from a gazetteer and/or perturbed versions of the named entities in the original utterances selected from a gazetteer. The perturbed versions of the named entities in the gazetteer can be generated with a perturbing function such as a typographical error generating function. The named entities in the original utterances can correspond to different named entity categories and a gazetteer can be generated for each named entity category by extracting the named entities corresponding to the respective categories in the original utterances and inserting them into the respective gazetteers for the respective categories. Each gazetteer can be expanded by searching a knowledge base for named entities that are similar to the named entities in the respective gazetteer and inserting the results of the search (i.e., the similar named entities) into the respective gazetteer. Additionally, each expanded gazetteer can be further expanded by applying the perturbing function to the named entities in the respective expanded gazetteers. Template utterances can be generated by removing the named entities from the original utterances and additional utterances can be generated by populating the template utterances with named entities selected from the gazetteers that are different from the named entities of the original utterances. The populated template utterances can be combined with the original utterances to form the augmented training data in which the NER model can be trained with.

In various embodiments, a computer-implemented method includes accessing training data comprising a plurality of original utterances, wherein each original utterance of the plurality of original utterances comprises at least one named entity corresponding to a named entity category of a plurality of named entity categories; accessing one or more gazetteers, wherein each gazetteer of the one or more gazetteers comprises a plurality of named entities extracted from the plurality of original utterances and a plurality of perturbed named entities derived from one or more named entities of the plurality of named entities; generating a plurality of template utterances, wherein each template utterance of the plurality of template utterances comprises information from an original utterance of the plurality of original utterances and at least one placeholder identifier representing a named entity in the original utterance of the plurality of original utterances; generating a plurality of additional utterances, wherein each additional utterance of the plurality of additional utterances comprises a template utterance of the plurality of template utterances populated with at least one named entity selected from a gazetteer of the one or more gazetteers; augmenting the training data by adding the plurality of additional utterances to the plurality of original utterances; and training a NER model with the augmented training data. Other features and advantages of the various embodiments are apparent throughout this disclosure.

Bot Systems

A bot (also referred to as a skill, chatbot, chatterbot, or talkbot) is a computer program that can perform conversations with end users. The bot can generally respond to natural-language messages (e.g., questions or comments) through a messaging application that uses natural-language messages. Enterprises may use one or more bots to communicate with end users through a messaging application. The messaging application may include, for example, over-the-top (OTT) messaging channels (such as Facebook Messenger, Facebook WhatsApp, WeChat, Line, Kik, Telegram, Talk, Skype, Slack, or SMS), virtual private assistants (such as Amazon Dot, Echo, or Show, Google Home, Apple HomePod, etc.), mobile and web app extensions that extend native or hybrid/responsive mobile apps or web applications with chat capabilities, or voice based input (such as devices or apps with interfaces that use Siri, Cortana, Google Voice, or other speech input for interaction).

In some examples, the bot may be associated with a Uniform Resource Identifier (URI). The URI may identify the bot using a string of characters. The URI may be used as a webhook for one or more messaging application systems. The URI may include, for example, a Uniform Resource Locator (URL) or a Uniform Resource Name (URN). The bot may be designed to receive a message (e.g., a hypertext transfer protocol (HTTP) post call message) from a messaging application system. The HTTP post call message may be directed to the URI from the messaging application system. In some examples, the message may be different from a HTTP post call message. For example, the bot may receive a message from a Short Message Service (SMS). While discussion herein refers to communications that the bot receives as a message, it should be understood that the message may be an HTTP post call message, a SMS message, or any other type of communication between two systems.

End users interact with the bot through conversational interactions (sometimes referred to as a conversational user interface (UI)), just as end users interact with other people. In some cases, the conversational interactions may include the end user saying "Hello" to the bot and the bot responding with a "Hi" and asking the end user how it can help. End users also interact with the bot through other types of interactions, such as transactional interactions (e.g., with a banking bot that is at least trained to transfer money from one account to another), informational interactions (e.g., with a human resources bot that is at least trained check the remaining vacation hours the user has), and/or retail interactions (e.g., with a retail bot that is at least trained for discussing returning purchased goods or seeking technical support).

In some examples, the bot may intelligently handle end user interactions without intervention by an administrator or developer of the bot. For example, an end user may send one or more messages to the bot in order to achieve a desired goal. A message may include certain content, such as text, emojis, audio, image, video, or other method of conveying a message. In some examples, the bot may automatically convert content into a standardized form and generate a natural language response. The bot may also automatically prompt the end user for additional input parameters or request other additional information. In some examples, the bot may also initiate communication with the end user, rather than passively responding to end user utterances.

A conversation with a bot may follow a specific conversation flow including multiple states. The flow may define what would happen next based on an input. In some examples, a state machine that includes user defined states (e.g., end user intents) and actions to take in the states or from state to state may be used to implement the bot. A conversation may take different paths based on the end user input, which may impact the decision the bot makes for the flow. For example, at each state, based on the end user input or utterances, the bot may determine the end user's intent in order to determine the appropriate next action to take. As used herein and in the context of an utterance, the term "intent" refers to an intent of the user who provided the utterance. For example, the user may intend to engage the bot in a conversation to order pizza, where the user's intent would be represented through the utterance "order pizza." A user intent can be directed to a particular task that the user wishes the bot to perform on behalf of the user. Therefore, utterances reflecting the user's intent can be phrased as questions, commands, requests, and the like.

In the context of the configuration of the bot, the term "intent" is also used herein to refer to configuration information for mapping a user's utterance to a specific task/action or category of task/action that the bot can perform. In order to distinguish between the intent of an utterance (i.e., a user intent) and the intent of the bot, the latter is sometimes referred to herein as a "bot intent." A bot intent may comprise a set of one or more utterances associated with the intent. For instance, an intent for ordering pizza can have various permutations of utterances that express a desire to place an order for pizza. These associated utterances can be used to train an intent classifier of the bot to enable the intent classifier to subsequently determine whether an input utterance from a user matches the order pizza intent. Bot intents may be associated with one or more dialog flows for starting a conversation with the user and in a certain state. For example, the first message for the order pizza intent could be the question "What kind of pizza would you like?" In addition to associated utterances, bot intents may further comprise named entities that relate to the intent. For example, the order pizza intent could include variables or parameters used to perform the task of ordering pizza (e.g., topping 1, topping 2, pizza type, pizza size, pizza quantity, and the like). The value of an entity is typically obtained through conversing with the user.

FIG. **1** is a simplified block diagram of an environment **100** incorporating a chatbot system according to certain embodiments. Environment **100** comprises a digital assistant builder platform (DABP) **102** that enables users **104** of DABP **102** to create and deploy digital assistants or chatbot systems. DABP **102** can be used to create one or more digital assistants (or DAs) or chatbot systems. For example, as shown in FIG. **1**, users **104** representing a particular enterprise can use DABP **102** to create and deploy a digital assistant **106** for users of the particular enterprise. For example, DABP **102** can be used by a bank to create one or more digital assistants for use by the bank's customers. The same DABP **102** platform can be used by multiple enterprises to create digital assistants. As another example, an owner of a restaurant (e.g., a pizza shop) may use DABP **102** to create and deploy a digital assistant that enables customers of the restaurant to order food (e.g., order pizza).

For purposes of this disclosure, a "digital assistant" is a tool that helps users of the digital assistant accomplish various tasks through natural language conversations. A digital assistant can be implemented using software only (e.g., the digital assistant is a digital tool implemented using programs, code, or instructions executable by one or more processors), using hardware, or using a combination of hardware and software. A digital assistant can be embodied or implemented in various physical systems or devices, such as in a computer, a mobile phone, a watch, an appliance, a vehicle, and the like. A digital assistant is also sometimes referred to as a chatbot system. Accordingly, for purposes of this disclosure, the terms digital assistant and chatbot system are interchangeable.

A digital assistant, such as digital assistant **106** built using DABP **102**, can be used to perform various tasks via natural language-based conversations between the digital assistant and its users **108**. As part of a conversation, a user may provide one or more user inputs **110** to digital assistant **106** and get responses **112** back from digital assistant **106**. A conversation can include one or more of inputs **110** and responses **112**. Via these conversations, a user can request one or more tasks to be performed by the digital assistant and, in response, the digital assistant is configured to perform the user-requested tasks and respond with appropriate responses to the user.

User inputs **110** are generally in a natural language form and are referred to as utterances. A user utterance **110** can be in text form, such as when a user types in a sentence, a question, a text fragment, or even a single word and provides it as input to digital assistant **106**. In some examples, a user utterance **110** can be in audio input or speech form, such as when a user says or speaks something that is provided as input to digital assistant **106**. The utterances are typically in a language spoken by the user. For example, the utterances may be in English, or some other language. When an utterance is in speech form, the speech input is converted to text form utterances in that particular language and the text utterances are then processed by digital assistant **106**. Various speech-to-text processing techniques may be used to convert a speech or audio input to a text utterance, which is then processed by digital assistant **106**. In some examples, the speech-to-text conversion may be done by digital assistant **106** itself.

An utterance, which may be a text utterance or a speech utterance, can be a fragment, a sentence, multiple sentences, one or more words, one or more questions, combinations of the aforementioned types, and the like. Digital assistant **106** is configured to apply natural language understanding (NLU) techniques to the utterance to understand the meaning of the user input. As part of the NLU processing for an utterance, digital assistant **106** is configured to perform processing to understand the meaning of the utterance, which involves identifying one or more intents and one or more entities corresponding to the utterance. Upon understanding the meaning of an utterance, digital assistant **106** may perform one or more actions or operations responsive to the understood meaning or intents. For purposes of this disclosure, it is assumed that the utterances are text utterances that have been provided directly by a user of digital assistant **106** or are the results of conversion of input speech utterances to text form. This however is not intended to be limiting or restrictive in any manner.

For example, a user input may request a pizza to be ordered by providing an utterance such as "I want to order a pizza." Upon receiving such an utterance, digital assistant **106** is configured to understand the meaning of the utterance and take appropriate actions. The appropriate actions may involve, for example, responding to the user with questions requesting user input on the type of pizza the user desires to order, the size of the pizza, any toppings for the pizza, and the like. The responses provided by digital assistant **106** may also be in natural language form and typically in the same language as the input utterance. As part of generating these responses, digital assistant **106** may perform natural language generation (NLG). For the user ordering a pizza, via the conversation between the user and digital assistant **106**, the digital assistant may guide the user to provide all the requisite information for the pizza order, and then at the end of the conversation cause the pizza to be ordered. Digital assistant **106** may end the conversation by outputting information to the user indicating that the pizza has been ordered.

At a conceptual level, digital assistant **106** performs various processing in response to an utterance received from a user. In some examples, this processing involves a series or pipeline of processing steps including, for example, understanding the meaning of the input utterance, determining an action to be performed in response to the utterance, where appropriate causing the action to be performed, generating a response to be output to the user responsive to the user utterance, outputting the response to the user, and the like. The NLU processing can include parsing the received input utterance to understand the structure and meaning of the utterance, refining, and reforming the utterance to develop a better understandable form (e.g., logical form) or structure for the utterance. Generating a response may include using NLG techniques.

The NLU processing performed by a digital assistant, such as digital assistant **106**, can include various NLP related tasks such as sentence parsing (e.g., tokenizing, lemmatizing, identifying part-of-speech tags for the sentence, identifying named entities in the sentence, generating dependency trees to represent the sentence structure, splitting a sentence into clauses, analyzing individual clauses, resolving anaphoras, performing chunking, and the like). In certain examples, the NLU processing is performed by digital assistant **106** itself. In some other examples, digital assistant **106** may use other resources to perform portions of the NLU processing. For example, the syntax and structure of an input utterance sentence may be identified by processing the sentence using a parser, a part-of-speech tagger, and/or a NER. In one implementation, for the English language, a parser, a part-of-speech tagger, and a named entity recognizer such as ones provided by the Stanford NLP Group are used for analyzing the sentence structure and syntax. These are provided as part of the Stanford CoreNLP toolkit.

While the various examples provided in this disclosure show utterances in the English language, this is meant only as an example. In certain examples, digital assistant **106** is also capable of handling utterances in languages other than English. Digital assistant **106** may provide subsystems (e.g., components implementing NLU functionality) that are configured for performing processing for different languages.

These subsystems may be implemented as pluggable units that can be called using service calls from an NLU core server. This makes the NLU processing flexible and extensible for each language, including allowing different orders of processing. A language pack may be provided for individual languages, where a language pack can register a list of subsystems that can be served from the NLU core server.

A digital assistant, such as digital assistant **106** depicted in FIG. **1**, can be made available or accessible to its users **108** through a variety of different channels, such as but not limited to, via certain applications, via social media platforms, via various messaging services and applications, and other applications or channels. A single digital assistant can have several channels configured for it so that it can be run on and be accessed by different services simultaneously.

A digital assistant or chatbot system generally contains or is associated with one or more skills. In certain embodiments, these skills are individual chatbots (referred to as skill bots) that are configured to interact with users and fulfill specific types of tasks, such as tracking inventory, submitting timecards, creating expense reports, ordering food, checking a bank account, making reservations, buying a widget, and the like. For example, for the embodiment depicted in FIG. **1**, digital assistant or chatbot system **106** includes skills **116-1**, **116-2**, **116-3**, and so on. For purposes of this disclosure, the terms "skill" and "skills" are used synonymously with the terms "skill bot" and "skill bots," respectively.

Each skill associated with a digital assistant helps a user of the digital assistant complete a task through a conversation with the user, where the conversation can include a combination of text or audio inputs provided by the user and responses provided by the skill bots. These responses may be in the form of text or audio messages to the user and/or using simple user interface elements (e.g., select lists) that are presented to the user for the user to make selections.

There are various ways in which a skill or skill bot can be associated or added to a digital assistant. In some instances, a skill bot can be developed by an enterprise and then added to a digital assistant using DABP **102**. In other instances, a skill bot can be developed and created using DABP **102** and then added to a digital assistant created using DABP **102**. In yet other instances, DABP **102** provides an online digital store (referred to as a "skills store") that offers multiple skills directed to a wide range of tasks. The skills offered through the skills store may also expose various cloud services. In order to add a skill to a digital assistant being generated using DABP **102**, a user of DABP **102** can access the skills store via DABP **102**, select a desired skill, and indicate that the selected skill is to be added to the digital assistant created using DABP **102**. A skill from the skills store can be added to a digital assistant as is or in a modified form (for example, a user of DABP **102** may select and clone a particular skill bot provided by the skills store, make customizations or modifications to the selected skill bot, and then add the modified skill bot to a digital assistant created using DABP **102**).

Various different architectures may be used to implement a digital assistant or chatbot system. For example, in certain embodiments, the digital assistants created and deployed using DABP **102** may be implemented using a master bot/child (or sub) bot paradigm or architecture. According to this paradigm, a digital assistant is implemented as a master bot that interacts with one or more child bots that are skill bots. For example, in the embodiment depicted in FIG. **1**, digital assistant **106** comprises a master bot **114** and skill bots **116-1**, **116-2**, etc. that are child bots of master bot **114**. In certain examples, digital assistant **106** is itself considered to act as the master bot.

A digital assistant implemented according to the master-child bot architecture enables users of the digital assistant to interact with multiple skills through a unified user interface, namely via the master bot. When a user engages with a digital assistant, the user input is received by the master bot. The master bot then performs processing to determine the meaning of the user input utterance. The master bot then determines whether the task requested by the user in the utterance can be handled by the master bot itself, else the master bot selects an appropriate skill bot for handling the user request and routes the conversation to the selected skill bot. This enables a user to converse with the digital assistant through a common single interface and still provide the capability to use several skill bots configured to perform specific tasks. For example, for a digital assistance developed for an enterprise, the master bot of the digital assistant may interface with skill bots with specific functionalities, such as a customer relationship management (CRM) bot for performing functions related to customer relationship management, an enterprise resource planning (ERP) bot for performing functions related to enterprise resource planning, a human capital management (HCM) bot for performing functions related to human capital management, etc. This way the end user or consumer of the digital assistant need only know how to access the digital assistant through the common master bot interface and behind the scenes multiple skill bots are provided for handling the user request.

In certain examples, in a master bot/child bots' infrastructure, the master bot is configured to be aware of the available list of skill bots. The master bot may have access to metadata that identifies the various available skill bots, and for each skill bot, the capabilities of the skill bot including the tasks that can be performed by the skill bot. Upon receiving a user request in the form of an utterance, the master bot is configured to, from the multiple available skill bots, identify or predict a specific skill bot that can best serve or handle the user request. The master bot then routes the utterance (or a portion of the utterance) to that specific skill bot for further handling. Control thus flows from the master bot to the skill bots. The master bot can support multiple input and output channels. In certain examples, routing may be performed with the aid of processing performed by one or more available skill bots. For example, as discussed below, a skill bot can be trained to infer an intent for an utterance and to determine whether the inferred intent matches an intent with which the skill bot is configured. Thus, the routing performed by the master bot can involve the skill bot communicating to the master bot an indication of whether the skill bot has been configured with an intent suitable for handling the utterance.

While the embodiment in FIG. **1** shows digital assistant **106** comprising a master bot **114** and skill bots **116-1**, **116-2**, and **116-3**, this is not intended to be limiting. A digital assistant can include various other components (e.g., other systems and subsystems) that provide the functionalities of the digital assistant. These systems and subsystems may be implemented only in software (e.g., code, instructions stored on a computer-readable medium and executable by one or more processors), in hardware only, or in implementations that use a combination of software and hardware.

DABP **102** provides an infrastructure and various services and features that enable a user of DABP **102** to create a digital assistant including one or more skill bots associated with the digital assistant. In some instances, a skill bot can be created by cloning an existing skill bot, for example, cloning a skill bot provided by the skills store. As previously indicated, DABP **102** provides a skills store or skills catalog that offers multiple skill bots for performing various tasks. A user of DABP **102** can clone a skill bot from the skills store. As needed, modifications or customizations may be made to the cloned skill bot. In some other instances, a user of DABP **102** created a skill bot from scratch using tools and services offered by DABP **102**. As previously indicated, the skills store or skills catalog provided by DABP **102** may offer multiple skill bots for performing various tasks.

In certain examples, at a high level, creating or customizing a skill bot involves the following steps:

(1) Configuring settings for a new skill bot
(2) Configuring one or more intents for the skill bot
(3) Configuring one or more entities for one or more intents
(4) Training the skill bot
(5) Creating a dialog flow for the skill bot
(6) Adding custom components to the skill bot as needed
(7) Testing and deploying the skill bot

Each of the above steps is briefly described below.

(1) Configuring settings for a new skill bot—Various settings may be configured for the skill bot. For example, a skill bot designer can specify one or more invocation names for the skill bot being created. These invocation names can then be used by users of a digital assistant to explicitly invoke the skill bot. For example, a user can input an invocation name in the user's utterance to explicitly invoke the corresponding skill bot.

(2) Configuring one or more intents and associated example utterances for the skill bot—The skill bot designer specifies one or more intents (also referred to as bot intents) for a skill bot being created. The skill bot is then trained based upon these specified intents. These intents represent categories or classes that the skill bot is trained to infer for input utterances. Upon receiving an utterance, a trained skill bot infers an intent for the utterance, where the inferred intent is selected from the predefined set of intents used to train the skill bot. The skill bot then takes an appropriate action responsive to an utterance based upon the intent inferred for that utterance. In some instances, the intents for a skill bot represent tasks that the skill bot can perform for users of the digital assistant. Each intent is given an intent identifier or intent name. For example, for a skill bot trained for a bank, the intents specified for the skill bot may include "CheckBalance," "TransferMoney," "DepositCheck," and the like.

For each intent defined for a skill bot, the skill bot designer may also provide one or more example utterances that are representative of and illustrate the intent. These example utterances are meant to represent utterances that a user may input to the skill bot for that intent. For example, for the CheckBalance intent, example utterances may include "What's my savings account balance?", "How much is in my checking account?", "How much money do I have in my account," and the like. Accordingly, various permutations of typical user utterances may be specified as example utterances for an intent.

The intents and their associated example utterances are used as training data to train the skill bot. Various different training techniques may be used. As a result of this training, a predictive model is generated that is configured to take an utterance as input and output an intent inferred for the utterance by the predictive model. In some instances, input utterances are provided to an intent analysis engine, which is configured to use the trained model to predict or infer an intent for the input utterance. The skill bot may then take one or more actions based upon the inferred intent.

(3) Configuring entities for one or more intents of the skill bot—In some instances, additional context may be needed to enable the skill bot to properly respond to a user utterance. For example, there may be situations where a user input utterance resolves to the same intent in a skill bot. For instance, in the above example, utterances "What's my savings account balance?" and "How much is in my checking account?" both resolve to the same CheckBalance intent, but these utterances are different requests asking for different things. To clarify such requests, one or more entities are added to an intent. Using the banking skill bot example, an entity called AccountType, which defines values called "checking" and "saving" may enable the skill bot to parse the user request and respond appropriately. In the above example, while the utterances resolve to the same intent, the value associated with the AccountType entity is different for the two utterances. This enables the skill bot to perform possibly different actions for the two utterances in spite of them resolving to the same intent. One or more entities can be specified for certain intents configured for the skill bot. Entities are thus used to add context to the intent itself. Entities help describe an intent more fully and enable the skill bot to complete a user request.

In certain examples, there are two types of entities: (a) built-in entities provided by DABP **102**, and (2) custom entities that can be specified by a skill bot designer. Built-in entities are generic entities that can be used with a wide variety of bots. Examples of built-in entities include, without limitation, entities related to time, date, addresses, numbers, email addresses, duration, recurring time periods, currencies, phone numbers, URLs, and the like. Custom entities are used for more customized applications. For example, for a banking skill, an AccountType entity may be defined by the skill bot designer that enables various banking transactions by checking the user input for keywords like checking, savings, and credit cards, etc.

(4) Training the skill bot—A skill bot is configured to receive user input in the form of utterances parse or otherwise process the received input and identify or select an intent that is relevant to the received user input. As indicated above, the skill bot has to be trained for this. In certain embodiments, a skill bot is trained based upon the intents configured for the skill bot and the example utterances associated with the intents (collectively, the training data), so that the skill bot can resolve user input utterances to one of its configured intents. In certain examples, the skill bot uses a predictive model that is trained using the training data and allows the skill bot to discern what users say (or in some cases, are trying to say). DABP **102** provides various different training techniques that can be used by a skill bot designer to train a skill bot, including various machine-learning based training techniques, rules-based training techniques, and/or combinations thereof. In certain examples, a portion (e.g., 80%) of the training data is used to train a skill bot model and another portion (e.g., the remaining 20%) is used to test or verify the model. Once trained, the trained model (also sometimes referred to as the trained skill bot) can then be used to handle and respond to user utterances. In certain cases, a user's utterance may be a question that requires only a single answer and no further conversation. In order to handle such situations, a Q&A (question-and-answer) intent may be defined for a skill bot. This enables a skill bot to output replies to user requests without having to update the dialog definition. Q&A intents

are created in a similar manner as regular intents. The dialog flow for Q&A intents can be different from that for regular intents.

(5) Creating a dialog flow for the skill bot—A dialog flow specified for a skill bot describes how the skill bot reacts as different intents for the skill bot are resolved responsive to received user input. The dialog flow defines operations or actions that a skill bot will take, e.g., how the skill bot responds to user utterances, how the skill bot prompts users for input, how the skill bot returns data. A dialog flow is like a flowchart that is followed by the skill bot. The skill bot designer specifies a dialog flow using a language, such as markdown language. In certain embodiments, a version of YAML called OBotML may be used to specify a dialog flow for a skill bot. The dialog flow definition for a skill bot acts as a model for the conversation itself, one that lets the skill bot designer choreograph the interactions between a skill bot and the users that the skill bot services.

In certain examples, the dialog flow definition for a skill bot contains three sections:

(a) a context sections

(b) a default transitions section

(c) states section

Context section—The skill bot designer can define variables that are used in a conversation flow in the context section. Other variables that may be named in the context section include, without limitation: variables for error handling, variables for built-in or custom entities, user variables that enable the skill bot to recognize and persist user preferences, and the like.

Default transitions section—Transitions for a skill bot can be defined in the dialog flow states section or in the default transitions section. The transitions defined in the default transition section act as a fallback and get triggered when there are no applicable transitions defined within a state, or the conditions required to trigger a state transition cannot be met. The default transitions section can be used to define routing that allows the skill bot to gracefully handle unexpected user actions.

States section—A dialog flow and its related operations are defined as a sequence of transitory states, which manage the logic within the dialog flow. Each state node within a dialog flow definition name a component that provides the functionality needed at that point in the dialog. States are thus built around the components. A state contains component-specific properties and defines the transitions to other states that get triggered after the component executes.

Special case scenarios may be handled using the states sections. For example, there might be times when you want to provide users the option to temporarily leave a first skill, they are engaged with to do something in a second skill within the digital assistant. For example, if a user is engaged in a conversation with a shopping skill (e.g., the user has made some selections for purchase), the user may want to jump to a banking skill (e.g., the user may want to ensure that he/she has enough money for the purchase), and then return to the shopping skill to complete the user's order. To address this, an action in the first skill can be configured to initiate an interaction with the second different skill in the same digital assistant and then return to the original flow.

(6) Adding custom components to the skill bot—As described above, states specified in a dialog flow for skill bot name components that provide the functionality needed corresponding to the states. Components enable a skill bot to perform functions. In certain embodiments, DABP **102** provides a set of preconfigured components for performing a wide range of functions. A skill bot designer can select one of more of these preconfigured components and associate them with states in the dialog flow for a skill bot. The skill bot designer can also create custom or new components using tools provided by DABP **102** and associate the custom components with one or more states in the dialog flow for a skill bot.

(7) Testing and deploying the skill bot—DABP **102** provides several features that enable the skill bot designer to test a skill bot being developed. The skill bot can then be deployed and included in a digital assistant.

While the description above describes how to create a skill bot, similar techniques may also be used to create a digital assistant (or the master bot). At the master bot or digital assistant level, built-in system intents may be configured for the digital assistant. These built-in system intents are used to identify general tasks that the digital assistant itself (i.e., the master bot) can handle without invoking a skill bot associated with the digital assistant. Examples of system intents defined for a master bot include: (1) Exit: applies when the user signals the desire to exit the current conversation or context in the digital assistant; (2) Help: applies when the user asks for help or orientation; and (3) Unresolved Intent: applies to user input that doesn't match well with the exit and help intents. The digital assistant also stores information about the one or more skill bots associated with the digital assistant. This information enables the master bot to select a particular skill bot for handling an utterance.

At the master bot or digital assistant level, when a user inputs a phrase or utterance to the digital assistant, the digital assistant is configured to perform processing to determine how to route the utterance and the related conversation. The digital assistant determines this using a routing model, which can be rules-based, AI-based, or a combination thereof. The digital assistant uses the routing model to determine whether the conversation corresponding to the user input utterance is to be routed to a particular skill for handling, is to be handled by the digital assistant or master bot itself per a built-in system intent or is to be handled as a different state in a current conversation flow.

In certain embodiments, as part of this processing, the digital assistant determines if the user input utterance explicitly identifies a skill bot using its invocation name. If an invocation name is present in the user input, then it is treated as explicit invocation of the skill bot corresponding to the invocation name. In such a scenario, the digital assistant may route the user input to the explicitly invoked skill bot for further handling. If there is no specific or explicit invocation, in certain embodiments, the digital assistant evaluates the received user input utterance and computes confidence scores for the system intents and the skill bots associated with the digital assistant. The score computed for a skill bot or system intent represents how likely the user input is representative of a task that the skill bot is configured to perform or is representative of a system intent. Any system intent or skill bot with an associated computed confidence score exceeding a threshold value (e.g., a Confidence Threshold routing parameter) is selected as a candidate for further evaluation. The digital assistant then selects, from the identified candidates, a particular system intent or a skill bot for further handling of the user input utterance. In certain embodiments, after one or more skill bots are identified as candidates, the intents associated with those candidate skills are evaluated (according to the intent model for each skill) and confidence scores are determined for each intent. In general, any intent that has a confidence score exceeding a threshold value (e.g., 70%) is treated as a candidate intent. If a particular skill bot is selected, then the user utterance is routed to that skill bot for further processing. If a system intent is selected, then one or more actions are performed by the master bot itself according to the selected system intent.

FIG. **2** is a simplified block diagram of a master bot (MB) system **200** according to certain embodiments. MB system **200** can be implemented in software only, hardware only, or a combination of hardware and software. MB system **200** includes a pre-processing subsystem **210**, a multiple intent subsystem (MIS) **220**, an explicit invocation subsystem (EIS) **230**, a skill bot invoker **240**, and a data store **250**. MB system **200** depicted in FIG. **2** is merely an example of an arrangement of components in a master bot. One of ordinary skill in the art would recognize many possible variations, alternatives, and modifications. For example, in some implementations, MB system **200** may have more or fewer systems or components than those shown in FIG. **2**, may combine two or more subsystems, or may have a different configuration or arrangement of subsystems.

Pre-processing subsystem **210** receives an utterance "A" **202** from a user and processes the utterance through a language detector **212** and a language parser **214**. As indicated above, an utterance can be provided in various ways including audio or text. The utterance **202** can be a sentence fragment, a complete sentence, multiple sentences, and the like. Utterance **202** can include punctuation. For example, if the utterance **202** is provided as audio, the pre-processing subsystem **210** may convert the audio to text using a speech-to-text converter (not shown) that inserts punctuation marks into the resulting text, e.g., commas, semicolons, periods, etc.

Language detector **212** detects the language of the utterance **202** based on the text of the utterance **202**. The manner in which the utterance **202** is handled depends on the language since each language has its own grammar and semantics. Differences between languages are taken into consideration when analyzing the syntax and structure of an utterance.

Language parser **214** parses the utterance **202** to extract part of speech (POS) tags for individual linguistic units (e.g., words) in the utterance **202**. POS tags include, for example, noun (NN), pronoun (PN), verb (VB), and the like. Language parser **214** may also tokenize the linguistic units of the utterance **202** (e.g., to convert each word into a separate token) and lemmatize words. A lemma is the main form of a set of words as represented in a dictionary (e.g., "run" is the lemma for run, runs, ran, running, etc.). Other types of pre-processing that the language parser **214** can perform include chunking of compound expressions, e.g., combining "credit" and "card" into a single expression "credit card." Language parser **214** may also identify relationships between the words in the utterance **202**. For example, in some embodiments, the language parser **214** generates a dependency tree that indicates which part of the utterance (e.g., a particular noun) is a direct object, which part of the utterance is a preposition, and so on. The results of the processing performed by the language parser **214** form extracted information **205** and are provided as input to MIS **220** together with the utterance **202** itself.

As indicated above, the utterance **202** can include more than one sentence. For purposes of detecting multiple intents and explicit invocation, the utterance **202** can be treated as a single unit even if it includes multiple sentences. However, in certain embodiments, pre-processing can be performed, e.g., by the pre-processing subsystem **210**, to identify a single sentence among multiple sentences for multiple intents analysis and explicit invocation analysis. In general, the results produced by MIS **220** and EIS **230** are substantially the same regardless of whether the utterance **202** is processed at the level of an individual sentence or as a single unit comprising multiple sentences.

MIS **220** determines whether the utterance **202** represents multiple intents. Although MIS **220** can detect the presence of multiple intents in the utterance **202**, the processing performed by MIS **220** does not involve determining whether the intents of the utterance **202** match to any intents that have been configured for a bot. Instead, processing to determine whether an intent of the utterance **202** matches a bot intent can be performed by an intent classifier **242** of the MB system **200** or by an intent classifier of a skill bot (e.g., as shown in FIG. **3**). The processing performed by MIS **220** assumes that there exists a bot (e.g., a particular skill bot or the master bot itself) that can handle the utterance **202**. Therefore, the processing performed by MIS **220** does not require knowledge of what bots are in the chatbot system (e.g., the identities of skill bots registered with the master bot), or knowledge of what intents have been configured for a particular bot.

To determine that the utterance **202** includes multiple intents, the MIS **220** applies one or more rules from a set of rules **252** in the data store **250**. The rules applied to the utterance **202** depend on the language of the utterance **202** and may include sentence patterns that indicate the presence of multiple intents. For example, a sentence pattern may include a coordinating conjunction that joins two parts (e.g., conjuncts) of a sentence, where both parts correspond to a separate intent. If the utterance **202** matches the sentence pattern, it can be inferred that the utterance **202** represents multiple intents. It should be noted that an utterance with multiple intents does not necessarily have different intents (e.g., intents directed to different bots or to different intents within the same bot). Instead, the utterance could have separate instances of the same intent (e.g., "Place a pizza order using payment account X, then place a pizza order using payment account Y").

As part of determining that the utterance **202** represents multiple intents, the MIS **220** also determines what portions of the utterance **202** are associated with each intent. MIS **220** constructs, for each intent represented in an utterance containing multiple intents, a new utterance for separate processing in place of the original utterance, e.g., an utterance "B" **206** and an utterance "C" **208**, as depicted in FIG. **2**. Thus, the original utterance **202** can be split into two or more separate utterances that are handled one at a time. MIS **220** determines, using the extracted information **205** and/or from analysis of the utterance **202** itself, which of the two or more utterances should be handled first. For example, MIS **220** may determine that the utterance **202** contains a marker word indicating that a particular intent should be handled first. The newly formed utterance corresponding to this particular intent (e.g., one of utterance **206** or utterance **208**) will be the first to be sent for further processing by EIS **230**. After a conversation triggered by the first utterance has ended (or has been temporarily suspended), the next highest priority utterance (e.g., the other one of utterance **206** or utterance **208**) can then be sent to the EIS **230** for processing.

EIS **230** determines whether the utterance that it receives (e.g., utterance **206** or utterance **208**) contains an invocation name of a skill bot. In certain embodiments, each skill bot in a chatbot system is assigned a unique invocation name that distinguishes the skill bot from other skill bots in the chatbot system. A list of invocation names can be maintained as part of skill bot information **254** in data store **250**. An utterance is deemed to be an explicit invocation when the utterance contains a word match to an invocation name. If a bot is not explicitly invoked, then the utterance received by the EIS **230** is deemed a non-explicitly invoking utterance **234** and is input to an intent classifier (e.g., intent classifier **242**) of the master bot to determine which bot to use for handling the utterance. In some instances, the intent classifier **242** will determine that the master bot should handle a non-explicitly invoking utterance. In other instances, the intent classifier **242** will determine a skill bot to route the utterance to for handling.

The explicit invocation functionality provided by the EIS **230** has several advantages. It can reduce the amount of processing that the master bot has to perform. For example, when there is an explicit invocation, the master bot may not have to do any intent classification analysis (e.g., using the intent classifier **242**), or may have to do reduced intent classification analysis for selecting a skill bot. Thus, explicit invocation analysis may enable selection of a particular skill bot without resorting to intent classification analysis.

Also, there may be situations where there is an overlap in functionalities between multiple skill bots. This may happen, for example, if the intents handled by the two skill bots overlap or are very close to each other. In such a situation, it may be difficult for the master bot to identify which of the multiple skill bots to select based upon intent classification analysis alone. In such scenarios, the explicit invocation disambiguates the particular skill bot to be used.

In addition to determining that an utterance is an explicit invocation, the EIS **230** is responsible for determining whether any portion of the utterance should be used as input to the skill bot being explicitly invoked. In particular, EIS **230** can determine whether part of the utterance is not associated with the invocation. The EIS **230** can perform this determination through analysis of the utterance and/or analysis of the extracted information **205**. EIS **230** can send the part of the utterance not associated with the invocation to the invoked skill bot in lieu of sending the entire utterance that was received by the EIS **230**. In some instances, the input to the invoked skill bot is formed simply by removing any portion of the utterance associated with the invocation. For example, "I want to order pizza using Pizza Bot" can be shortened to "I want to order pizza" since "using Pizza Bot" is relevant to the invocation of the pizza bot, but irrelevant to any processing to be performed by the pizza bot. In some instances, EIS **230** may reformat the part to be sent to the invoked bot, e.g., to form a complete sentence. Thus, the EIS **230** determines not only that there is an explicit invocation, but also what to send to the skill bot when there is an explicit invocation. In some instances, there may not be any text to input to the bot being invoked. For example, if the utterance was "Pizza Bot," then the EIS **230** could determine that the pizza bot is being invoked, but there is no text to be processed by the pizza bot. In such scenarios, the EIS **230** may indicate to the skill bot invoker **240** that there is nothing to send.

Skill bot invoker **240** invokes a skill bot in various ways. For instance, skill bot invoker **240** can invoke a bot in response to receiving an indication **235** that a particular skill bot has been selected as a result of an explicit invocation. The indication **235** can be sent by the EIS **230** together with the input for the explicitly invoked skill bot. In this scenario, the skill bot invoker **240** will turn control of the conversation over to the explicitly invoked skill bot. The explicitly invoked skill bot will determine an appropriate response to the input from the EIS **230** by treating the input as a stand-alone utterance. For example, the response could be to perform a specific action or to start a new conversation in a particular state, where the initial state of the new conversation depends on the input sent from the EIS **230**.

Another way in which skill bot invoker **240** can invoke a skill bot is through implicit invocation using the intent classifier **242**. The intent classifier **242** can be trained, using machine-learning and/or rules-based training techniques, to determine a likelihood that an utterance is representative of a task that a particular skill bot is configured to perform. The intent classifier **242** is trained on different classes, one class for each skill bot. For instance, whenever a new skill bot is registered with the master bot, a list of example utterances associated with the new skill bot can be used to train the intent classifier **242** to determine a likelihood that a particular utterance is representative of a task that the new skill bot can perform. The parameters produced as result of this training (e.g., a set of values for parameters of a machine-learning model) can be stored as part of skill bot information **254**.

In certain embodiments, the intent classifier **242** is implemented using a machine-learning model, as described in further detail herein. Training of the machine-learning model may involve inputting at least a subset of utterances from the example utterances associated with various skill bots to generate, as an output of the machine-learning model, inferences as to which bot is the correct bot for handling any particular training utterance. For each training utterance, an indication of the correct bot to use for the training utterance may be provided as ground truth information. The behavior of the machine-learning model can then be adapted (e.g., through back-propagation) to minimize the difference between the generated inferences and the ground truth information.

In certain embodiments, the intent classifier **242** determines, for each skill bot registered with the master bot, a confidence score indicating a likelihood that the skill bot can handle an utterance (e.g., the non-explicitly invoking utterance **234** received from EIS **230**). The intent classifier **242** may also determine a confidence score for each system level intent (e.g., help, exit) that has been configured. If a particular confidence score meets one or more conditions, then the skill bot invoker **240** will invoke the bot associated with the particular confidence score. For example, a threshold confidence score value may need to be met. Thus, an output **245** of the intent classifier **242** is either an identification of a system intent or an identification of a particular skill bot. In some embodiments, in addition to meeting a threshold confidence score value, the confidence score must exceed the next highest confidence score by a certain win margin. Imposing such a condition would enable routing to a particular skill bot when the confidence scores of multiple skill bots each exceed the threshold confidence score value.

After identifying a bot based on evaluation of confidence scores, the skill bot invoker **240** hands over processing to the identified bot. In the case of a system intent, the identified bot is the master bot. Otherwise, the identified bot is a skill bot. Further, the skill bot invoker **240** will determine what to provide as input **247** for the identified bot. As indicated above, in the case of an explicit invocation, the input **247** can be based on a part of an utterance that is not associated with the invocation, or the input **247** can be nothing (e.g., an empty string). In the case of an implicit invocation, the input **247** can be the entire utterance.

Data store **250** comprises one or more computing devices that store data used by the various subsystems of the master bot system **200**. As explained above, the data store **250** includes rules **252** and skill bot information **254**. The rules **252** include, for example, rules for determining, by MIS **220**, when an utterance represents multiple intents and how to split an utterance that represents multiple intents. The rules **252** further include rules for determining, by EIS **230**, which parts of an utterance that explicitly invokes a skill bot to send to the skill bot. The skill bot information **254** includes invocation names of skill bots in the chatbot system, e.g., a list of the invocation names of all skill bots registered with a particular master bot. The skill bot information **254** can also include information used by intent classifier **242** to determine a confidence score for each skill bot in the chatbot system, e.g., parameters of a machine-learning model.

FIG. **3** is a simplified block diagram of a skill bot system **300** according to certain embodiments. Skill bot system **300** is a computing system that can be implemented in software only, hardware only, or a combination of hardware and software. In certain embodiments such as the embodiment depicted in FIG. **1**, skill bot system **300** can be used to implement one or more skill bots within a digital assistant.

Skill bot system **300** includes an MIS **310**, an intent classifier **320**, and a conversation manager **330**. The MIS **310** is analogous to the MIS **220** in FIG. **2** and provides similar functionality, including being operable to determine, using rules **352** in a data store **350**: (1) whether an utterance represents multiple intents and, if so, (2) how to split the utterance into a separate utterance for each intent of the multiple intents. In certain embodiments, the rules applied by MIS **310** for detecting multiple intents and for splitting an utterance are the same as those applied by MIS **220**. The MIS **310** receives an utterance **302** and extracted information **304**. The extracted information **304** is analogous to the extracted information **205** in FIG. **1** and can be generated using the language parser **214** or a language parser local to the skill bot system **300**.

Intent classifier **320** can be trained in a similar manner to the intent classifier **242** discussed above in connection with the embodiment of FIG. **2** and as described in further detail herein. For instance, in certain embodiments, the intent classifier **320** is implemented using a machine-learning model. The machine-learning model of the intent classifier **320** is trained for a particular skill bot, using at least a subset of example utterances associated with that particular skill bot as training utterances. The ground truth for each training utterance would be the particular bot intent associated with the training utterance.

The utterance **302** can be received directly from the user or supplied through a master bot. When the utterance **302** is supplied through a master bot, e.g., as a result of processing through MIS **220** and EIS **230** in the embodiment depicted in FIG. **2**, the MIS **310** can be bypassed so as to avoid repeating processing already performed by MIS **220**. However, if the utterance **302** is received directly from the user, e.g., during a conversation that occurs after routing to a skill bot, then MIS **310** can process the utterance **302** to determine whether the utterance **302** represents multiple intents. If so, then MIS **310** applies one or more rules to split the utterance **302** into a separate utterance for each intent, e.g., an utterance "D" **306** and an utterance "E" **308**. If utterance **302** does not represent multiple intents, then MIS **310** forwards the utterance **302** to intent classifier **320** for intent classification and without splitting the utterance **302**.

Intent classifier **320** is configured to match a received utterance (e.g., utterance **306** or **308**) to an intent associated with skill bot system **300**. As explained above, a skill bot can be configured with one or more intents, each intent including at least one example utterance that is associated with the intent and used for training a classifier. In the embodiment of FIG. **2**, the intent classifier **242** of the master bot system **200** is trained to determine confidence scores for individual skill bots and confidence scores for system intents. Similarly, intent classifier **320** can be trained to determine a confidence score for each intent associated with the skill bot system **300**. Whereas the classification performed by intent classifier **242** is at the bot level, the classification performed by intent classifier **320** is at the intent level and therefore finer grained. The intent classifier **320** has access to intents information **354**. The intents information **354** includes, for each intent associated with the skill bot system **300**, a list of utterances that are representative of and illustrate the meaning of the intent and are typically associated with a task performable by that intent. The intents information **354** can further include parameters produced as a result of training on this list of utterances.

Conversation manager **330** receives, as an output of intent classifier **320**, an indication **322** of a particular intent, identified by the intent classifier **320**, as best matching the utterance that was input to the intent classifier **320**. In some instances, the intent classifier **320** is unable to determine any match. For example, the confidence scores computed by the intent classifier **320** could fall below a threshold confidence score value if the utterance is directed to a system intent or an intent of a different skill bot. When this occurs, the skill bot system **300** may refer the utterance to the master bot for handling, e.g., to route to a different skill bot. However, if the intent classifier **320** is successful in identifying an intent within the skill bot, then the conversation manager **330** will initiate a conversation with the user.

The conversation initiated by the conversation manager **330** is a conversation specific to the intent identified by the intent classifier **320**. For instance, the conversation manager **330** may be implemented using a state machine configured to execute a dialog flow for the identified intent. The state machine can include a default starting state (e.g., for when the intent is invoked without any additional input) and one or more additional states, where each state has associated with it actions to be performed by the skill bot (e.g., executing a purchase transaction) and/or dialog (e.g., questions, responses) to be presented to the user. Thus, the conversation manager **330** can determine an action/dialog **335** upon receiving the indication **322** identifying the intent and can determine additional actions or dialog in response to subsequent utterances received during the conversation.

Data store **350** comprises one or more computing devices that store data used by the various subsystems of the skill bot system **300**. As depicted in FIG. **3**, the data store **350** includes the rules **352** and the intents information **354**. In certain embodiments, data store **350** can be integrated into a data store of a master bot or digital assistant, e.g., the data store **250** in FIG. **2**.

Training Data Augmentation

It has been discovered that many NER models appear to pay more attention to the entity values in an utterance than to the contextual information of utterance. When an NER model pays more attention to the entity values than to the contextual information, the NER model does not generalize well to variations of entities and new and unseen entities in input utterances. To restate, contextual information helps the NER model perform well on utterances that include entities that are not in the training data and entities that are misspelled, spelled differently, or otherwise vary in some aspect with respect to an entity value in the training data. In order to overcome these problems, various embodiments are directed to techniques for augmenting training data using gazetteers and perturbations. The training data includes original utterances with each original utterance including at least one named entity corresponding to a named entity category. The training data can be augmented by generating additional utterances from the original utterances in the training data using gazetteers and perturbations and combining the generated additional utterances with the original utterances to form the augmented training data. A NER model can be trained with the augmented training data, which results in improved performance over NER models trained with just the original training data. Using the training data augmented as described herein, a NER model can learn to consider context in detecting and classifying named entities in utterances even when the named entities in the utterances do not match the named entities in the training data. Additionally, the NER model can learn to detect and classify named entities in the utterances even if the named entities in the utterances include typographical errors. In this way, a NER model trained using the training data augmented with the techniques described herein can generalize well to variations of named entities and new and unseen named entities. The NER model trained with the training data augmented with the techniques described herein may be implemented in a chatbot system, such as the chatbot system described with respect to FIGS. **1**, **2**, and **3**. Advantageously, these models and chatbots will perform better on utterances in various text formats because the models are better able to recognize and extract entities within the utterances.

FIG. **4**A shows a block diagram illustrating aspects of a model system **400** configured to train and provide one or more models, e.g., implemented as a NER model, based on text data. As shown in FIG. **4**A, the model system **400** in this example includes various subsystems: a prediction model training subsystem **410** to build and train models, an evaluation subsystem **415** to evaluate performance of trained models, and an implementation subsystem **420** for implementing the models as an artificial intelligence-based solution (e.g., deployed as part of a one or more chatbots). The prediction model training subsystem **410** builds and trains one or more prediction models $\mathbf{425}_A$-$\mathbf{425}_N$ ('N' represents any natural number) to be used by the other subsystems (which may be referred to herein individually as a prediction model **425** or collectively as the prediction models **425**). For example, the prediction models **425** can include a model for recognizing one or more entities in an utterance (e.g., a NER model), another model for determining a likelihood that an utterance is representative of a task that a particular skill bot is configured to perform, another model for predicting an intent from an utterance for a first type of skill bot, and another model for predicting an intent from an utterance for a second type of skill bot. Still other types of prediction models may be implemented in other examples according to this disclosure.

A prediction model **425** can be a machine-learning ("ML") model, such as a convolutional neural network ("CNN"), e.g. an inception neural network, a residual neural network ("Resnet"), or a recurrent neural network, e.g., long short-term memory ("LSTM") models or gated recurrent units ("GRUs") models, other variants of Deep Neural Networks ("DNN") (e.g., a multi-label n-binary DNN classifier or multi-class DNN classifier for single intent classification. A prediction model **425** can also be any other suitable ML model trained for natural language processing, such as a Naive Bayes Classifier, Linear Classifier, Support Vector Machine, Bagging Models such as Random Forest Model, Boosting Models, Shallow Neural Networks, or combinations of one or more of such techniques—e.g., CNN-HMM or MCNN (Multi-Scale Convolutional Neural Network). The chatbot system **400** may employ the same type of prediction model **425** or different types of prediction models **425** for recognizing one or more entities in an utterance, determining a likelihood that an utterance is representative of a task that a particular skill bot is configured to perform, predicting an intent from an utterance for a first type of skill bot, and predicting an intent from an utterance for a second type of skill bot.

To train the various prediction models **425**, the prediction model training subsystem **410** is comprised of three main components: dataset preparation **430**, data augmentation **435**, and model training **440**. The dataset preparation **430** includes the process of loading data assets **445**, splitting the data assets **445** into training and validation sets $\mathbf{445}_{A-N}$ so that the system can train and test the prediction models **425**, and performing basic pre-processing. The splitting the data assets **445** into training and validation sets $\mathbf{445}_{A-N}$ may be performed randomly (e.g., a 90/10% or 70/30%) or the splitting may be performed in accordance with a more complex validation technique such as K-Fold Cross-Validation, Leave-one-out Cross-Validation, Leave-one-group-out Cross-Validation, Nested Cross-Validation, or the like to minimize sampling bias and overfitting.

The training data may include at least a subset of utterances from example utterances associated with one or more skill bots. As indicated above, an utterance can be provided in various ways including audio or text. The utterance can be a sentence fragment, a complete sentence, multiple sentences, and the like. For example, if the utterance is provided as audio, the data preparation **430** may convert the audio to text using a speech-to-text converter (not shown) that inserts punctuation marks into the resulting text, e.g., commas, semicolons, periods, etc. In some instances, the example utterances are provided by a client or customer. In other instances, the example utterances are automatically generated from prior libraries of utterances (e.g., identifying utterances from a library that are specific to a skill that a chatbot is designated to learn). The training data for the prediction model **425** can include input text or audio (or input features of text or audio frames) and labels **450** corresponding to the input text or audio (or input features) as a matrix or table of values. For example, for each training utterance, an indication of the correct entities and classification thereof to be inferred by the prediction model **425** may be provided as ground truth information for labels **450**. The behavior of the prediction model **425** can then be adapted (e.g., through back-propagation) to minimize the difference between the generated inferences for various entities and the ground truth information.

The data augmentation **435** includes the process of augmenting the training data to include generated additional utterances that make the predictive model **425** more resilient towards utterances with variations of entities and new and unseen entities. By augmenting the training data with such generated additional utterances, the predictive model **425** becomes better at considering context in detecting and classifying entities in utterances even when the named entities in utterances do not match the named entities in the training data. Additionally, the predictive model **425** can learn to detect and classify entities in the utterances even if the entities in the utterances include typographical errors and/or are otherwise varied in their spelling.

The additional utterances are generated from the original utterances in the training data by replacing the named entities in the original utterances with different named entities selected from a gazetteer and/or with perturbed versions of the named entities in the original utterances selected from a gazetteer. The perturbed versions of the named entities in the gazetteer can be generated with a perturbing function such as a typographical error generating function. The named entities in the original utterances can correspond to different named entity categories and a gazetteer can be generated for each named entity category by extracting the named entities corresponding to the respective categories in the original utterances and inserting them into the respective gazetteers for the respective categories. Each gazetteer can be expanded by searching a knowledge base for named entities that are similar to the named entities in the respective gazetteer and inserting the results of the search (i.e., the similar named entities) into the respective gazetteer. Additionally, each expanded gazetteer can be further expanded by applying the typographical error generating function to the named entities in the respective expanded gazetteers. Template utterances can be generated by removing the named entities from the original utterances and additional utterances can be generated by populating the template utterances with named entities selected from the gazetteers that are different from the named entities of the original utterances. The populated template utterances can be combined with the original utterances to form the augmented training data in which the NER model can be trained with. Gazetteer generation, perturbation, and templates are described in more detail below.

The training process for the prediction model **425** includes selecting hyperparameters for the prediction model **425** and performing iterative operations of inputting utterances from the augmented training data into the prediction model **425** to find a set of model parameters (e.g., weights and/or biases) that maximizes or minimizes an objective function, e.g., minimizes a loss function, for the prediction model **425**. The hyperparameters are settings that can be tuned or optimized to control the behavior of the prediction model **425**. Most models explicitly define hyperparameters that control different aspects of the models such as memory or cost of execution. However, additional hyperparameters may be defined to adapt a model to a specific scenario. For example, the hyperparameters may include the number of hidden units or layers of a model, the learning rate of a model, the convolution kernel width, or the number of parameters for a model. The objective function can be constructed to measure the difference between the outputs inferred using the prediction models **425** and the ground truth annotated to the samples using the labels **450**. For example, for a supervised learning-based model, the goal of the training is to learn a function "h( )" (also sometimes referred to as the hypothesis function) that maps the training input space X to the target value space Y, h: X→Y, such that h(x) is a good predictor for the corresponding value of y. Various different techniques may be used to learn this hypothesis function. In some techniques, as part of deriving the hypothesis function, the objective function may be defined that measures the difference between the ground truth value for input and the predicted value for that input. As part of training, techniques such as back propagation, random feedback, Direct Feedback Alignment (DFA), Indirect Feedback Alignment (IFA), Hebbian learning, and the like are used update the model parameters in such a manner as to minimize or maximize this objective function.

Once the set of model parameters are identified, the prediction model **425** has been trained and can be tested or validated using the subset of testing data (testing or validation data set). The testing or validation process includes iterative operations of inputting utterances from the subset of testing data into the prediction model **425** using a validation technique such as K-Fold Cross-Validation, Leave-one-out Cross-Validation, Leave-one-group-out Cross-Validation, Nested Cross-Validation, or the like to tune the hyperparameters and ultimately find the optimal set of hyperparameters. Once the optimal set of hyperparameters are obtained, a reserved test set from the subset of test data may be input into the prediction model **425** to obtain output (in this example, one or more recognized entities), and the output is evaluated versus ground truth entities using correlation techniques such as Bland-Altman method and the Spearman's rank correlation coefficients. Further, performance metrics **452** may be calculated in entity recognition evaluation subsystem **415** such as the error, accuracy, precision, recall, receiver operating characteristic curve (ROC), etc. The performance metrics **452** may be used in the entity recognition evaluation subsystem **415** to analyze performance of the prediction model **425** for recognizing entities.

The model training subsystem **440** outputs trained models including one or more NER models **455**. The one or more NER models **455** may be deployed and used in the prediction model training subsystem **410** and/or implementation subsystem **420** for implementing an artificial intelligence-based solution. For example, the NER models **455** may be used to predict named entities **465** for the utterances in the training data and generate labels for the predicted named entities **465**. The training data can be used for further training the one or more NER models **455** and/or other prediction models **425** such as embedding models and/or intent prediction models. Alternatively, an artificial intelligence-based solution such as a chatbot may be configured with the one or more NER models **455** to receive text data **460**, including utterances, from one or more users and recognize and extract entities **465** from various utterances received by the one or more chatbots. The entities corresponding to the entity predictions **465** may be part of the extracted information (e.g., extracted information **205**; **304** described respectively in FIGS. **2** and **3**) obtained from the text data **460**, and may be used in downstream processing such as dialogue construction, data retrieval, or intent classification.

As discussed above, intent prediction and entity extraction help chatbot systems understand user queries and user utterances with respect to the domain of a given service or set of services. Intent prediction determines an objective (i.e., intent) of the user's query or utterance. Entity extraction determines one or more entities (i.e., constraints), if any, of the user's query or utterance. For example, for a user inquiry regarding "the weather on Wednesday in the Poconos," intent prediction determines that the user's intent is to learn about the "weather" and entity extraction determines that "Wednesday" and "Poconos" are entities that focus the user's intent to a particular day and geographical location. Entity detection and classification can be accomplished using a NER model. An entity label can be assigned to each instance of a named entity within the user utterance. In some examples, the named entities can correspond to named entity categories such as locations, organizations, times, persons, etc. For a given utterance, the words of the utterance that refer to an entity within a category of entities can be labeled with a label that identifies the category. For example, for the utterance "My name is Jane Doe," the words "Jane Doe" can be detected and classified as corresponding to a person and can be labeled with a label that identifies the category to which they have been classified (e.g., [$_{PER}$ Jane Doe]). Since the words "My name is" are contextual, they would not be detected and classified as named entities and a label that identifies those words as a named entity would not be assigned to them. In another example, for the utterance "What is the weather in Paris," the word "Paris" would be detected and classified as corresponding to a location and would be labeled as "[$_{LOC}$ Paris]." An utterance can include named entities that belong to different named entity categories and each named entity in the utterance can be detected, classified according to its named entity category, and labeled with a label that identifies its respective category. For example, for the utterance, "Her name is Jane Doe and she lives in Paris," the words "Jane Doe" can be detected, classified, and labeled as [$_{PER}$ Jane Doe] and the word "Paris" can be detected, classified, and labeled as [$_{LOC}$ Paris].

In order to label named entities, an NER model is typically trained with training data that includes every category of entity desired to be classified. For example, for an NER model to detect and classify the words "Jane Doe" as being a named entity that corresponds to a person and "Paris" as being a named entity that correspond to a location, the NER model would need to be trained with training data that incorporates named entities in the person category of named entities and named entities in a location category of named entities. Training data is typically obtained from public or privately pre-labeled data sets. Several datasets are available for training purposes, which includes text with labeled named entities corresponding to persons, locations, organizations, and names of miscellaneous entities that do not belong to the previous three groups. An example of a publicly available dataset in which the training data can be obtained from is the CoNLL-2003 dataset. Additional information for the ConLL-2003 dataset is found in "Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition" by Sang et al., published in arXiv preprint arXiv:cs/0306050, the entire contents of which are hereby incorporated by reference as if fully set forth herein.

The problem with public or privately pre-labeled datasets though, is that these datasets do not include pre-labeled data that is diverse enough to train the models to detect entities with all kinds of context and entity value variations (currency types, number formats, data time formats, different spelling conventions, etc.). Thus, the NER models trained with these pre-labeled datasets and other similar pre-labeled datasets typically learn to focus on the named entity values that they are trained on while ignoring contextual information. As a result, NER models have difficulty identifying and labelling named entities that the NER model is not specifically trained to identify and label (i.e., named entities that are not in the training data and/or named entities that are spelled differently, misspelled, vary, etc.). Previous attempts to address this problem have included generating customized labeled datasets (e.g., by crowd workers) or a user or customer of the NER model; however, generating these customized labeled datasets utilizes significant resources (e.g., human time and financial resources). To overcome these challenges and others, the approaches described herein utilizes gazetteers and perturbations to automatically augment the training data in less time and with less resources.

Gazetteers and Perturbations

The process for augmenting the training data begins with obtaining the training data and generating gazetteers. The training data includes utterances. Utterances in the obtained training data can be referred to as original utterances. Each utterance in the obtained training data can include one or more named entities corresponding to one or more named entity categories of a set of named entity categories. In some implementations, the set of named entity categories can correspond to default or system named entities (e.g., PERSON, NUMBER, CURRENCY, DATE_TIME). In other implementations, the set of named entity categories can be defined by a user, a customer, a developer, and the like, of a NER model. Each named entity in an utterance can include a label that indicates the category to which the named entity belongs. Each category of the set of named entity categories can be associated with a different label. For example, a person named entity category can be associated with a [$_{PER}$] label and a location named entity category can be associated with a [$_{LOC}$]. Each utterance in the training data can also include contextual information before and/or after the one or more named entities in the utterance. For example, for the utterance "His name is [$_{PER}$ John Doe]," the "His name is" portion of the utterance is contextual information that is before the named entity [$_{PER}$ John Doe]. The training data can be obtained from one or more public or privately pre-labeled datasets such as the CoNLL-2003 dataset described above.

A gazetteer can be generated for each named entity category of the set of named entity categories. As used herein, a gazetteer refers to a data structure that stores all of the named entities in the utterances of the training data for a particular category of the set of named entity categories. A gazetteer can be generated for a named entity category by extracting the named entities from the utterances of the training data that correspond to the respective named entity category and inserting the extracted named entities into the gazetteer for the respective category. For example, all of the named entities in the original utterances that correspond to the person category can be extracted and inserted into a person gazetteer and all of the named entities in the original utterances that correspond to the location category can be extracted and inserted into a location gazetteer. Specifically, continuing with the above example, for original utterances in the training data "My name is [$_{PER}$ Jane Doe]" and "His name is [$_{PER}$ John Doe]", an example of a gazetteer for the person category is $GZ_{[PER]}$: {[Jane Doe]; [John Doe]; . . . }. Similarly, continuing with the example, for original utterances "She lives in [$_{LOC}$ Paris]" and "He lives in [$_{LOC}$ New Yor]," an example of a gazetteer for the location category is $GZ_{[LOC]}$: {[Paris]; [New York]; . . . }.

Each respective gazetteer can be expanded by searching one or more knowledge bases for named entities that are similar to the named entities in the respective gazetteer and inserting the results of the search (i.e., the similar named entities) in the respective gazetteer. For example, the person gazetteer $GZ_{[PER]}$ described above can be expanded by searching a knowledge base such as Wikipedia for named entities that are similar to the named entities [Jane Doe] and [John Doe] in the person gazetteer $GZ_{[PER]}$ (i.e., those extracted from the utterances of the training data) and inserting the named entities that are found as a result of the search (e.g., [Joe Doe]) in the person gazetteer $GZ_{[PER]}$ (e.g., $GZ_{[PER]}$: {[Jane Doe]; [John Doe]; [Joe Doe] . . . }).

The one or more knowledge bases to be searched can include any publicly and/or privately available knowledge base or combination of publicly available and/or privately available knowledge bases. Examples of publicly available knowledge bases include but are not limited to Wikipedia, Wikidata, Wikimedia, DBpedia, and the like. A knowledge base can be searched by searching one or more publicly and/or privately available dump files (i.e., record files, archival files, etc.) for the knowledge base using a pre-trained model.

The pre-trained model can be configured to receive a gazetteer and one or more dump files for a knowledge base or knowledge bases to be searched as inputs, retrieve a set of retrieved named entities by searching the one or more dump files for named entities that are similar to the named entities in the gazetteer, processing the set of retrieved named entities to produce a set of processed named entities, and combining the set of processed named entities with the named entities in the gazetteer. For example, the pre-trained model can receive, as inputs, a person gazetteer (e.g., $GZ_{[PER]}$: {[Jane Doe]; [John Doe]; . . . }) and one or more Wikipedia dump files, retrieve a set of retrieved named entities (e.g., [Jayne Doe], [Jon Doe], [John Doe], [Jane Doe], . . .) by searching the one or more Wikipedia dump files for named entities that are similar to the named entities in the person gazetteer, processing the set of retrieved named entities to produce a set of processed named entities (e.g., [Jayne Doe], [Jon Doe], . . .), and combining the set of processed name entities with the named entities in the person gazetteer (e.g., $GZ_{[PER]}$: {[Jane Doe]; [John Doe]; [Jayne Doe]; [Jon Doe]; . . . }).

In some implementations, the pre-trained model can be the Wikipedia2Vec model. The Wikipedia2Vec model is a part of an Wikipedia2Vec open-source tool that can identify named entities in Wikipedia content from Wikipedia dump files and retrieve identified named entities by querying similar entities. Additional information for the Wikipedia2Vec model and tool is found in "Wikipedia2Vec: An Efficient Toolkit for Learning and Visualizing the Embeddings of Words and Entities from Wikipedia" by Yamada et al., published in arXiv preprint arXiv: 1812.06280, the entire contents of which are hereby incorporated by reference as if fully set forth herein.

In some implementations, a set of similarity scores can be calculated for the set of retrieved named entities. In other words, a similarity score can be calculated for each retrieved named entity in the set of retrieved named entities. The calculated score for a retrieved named entity can represent how similar the retrieved named entity is to any of the named entities in the gazetteer received as an input to pre-trained model. The calculated score can range between 0 and 1, with 0 indicating that there is a no similarity between the retrieved named entity and a named entity in the gazetteer and 1 indicating that the retrieved named entity is identical to a named entity in the gazetteer. In some implementations, retrieved named entities having a similarity greater than or equal to a predetermined similarity threshold can be included in the set of the retrieved named entities.

In some implementations, the set of retrieved named entities can be processed to produce a set of processed named entities by removing from the set of retrieved named entities artifacts (e.g., parentheticals, punctuation, hyphens, etc.), any named entities that are a combination of digits and symbols (i.e., no letters), and any named entities that are duplicative (i.e., either identical to a named entity in a gazetteer or identical to another named entity in the set of retrieved named entities). In some implementations, the case of each named entity in the set of retrieved named entities can be converted to lowercase, uppercase, and/or to match the case of the named entities in the gazetteer.

Additionally, or alternatively, each respective gazetteer can be further expanded by querying one or more knowledge bases for named entities that are categorized into one or more classes that are similar to the named entity category for the respective gazetteer. For example, the person gazetteer $GZ_{[PER]}$ described above can be further expanded by querying Wikipedia for named entities that are categorized into one or more classes (e.g., people, figures) that are similar to the named entity category (e.g., person) for the person gazetteer $GZ_{[PER]}$ and inserting the named entities that are found as a result of the query (e.g., [Joan Doe], [Jonathan Doe]) in the person gazetteer $GZ_{[PER]}$ (e.g., $GZ_{[PER]}$: {[Jane Doe]; [John Doe]; [Joan Doe]; [Jonathan Doe]; . . . }). A knowledge base can be queried by querying the one or more publicly available dump files for the knowledge base using a query-based search.

In order to further expand a respective gazetteer, a query-based search can be performed on the one or more dump files for a knowledge base or knowledge bases to be queried to retrieve a set of retrieved named entities that are categorized into one or more sub-classes and/or classes that are similar to the named entity category of the respective gazetteer.

Knowledge base data structures often store the content data of the knowledge base by named entity and the sub-category or category (i.e., property) to which the named entity belongs. For example, a data structure can store, in a table, the named entity and an identifier for the named entity in a column, a sub-class or class for the named entity and an identifier for the sub-class or class in another column, and a value that describes the sub-class or class and an identifier for the value in another column. In these knowledge base data structures, named entities are assigned a sub-category or category based on the properties of the named entities and sub-categories are assigned a category based on the properties of the named entities and the sub-categories. For example, for a knowledge base including knowledge about the city of Boston, the named entity Boston would be assigned the sub-category City because Boston is an instance of a city (i.e., Boston is a sub-category of a city). Similarly, Boston and City would be assigned the category State because Boston is an instance of a city in the state of Massachusetts (i.e., Boston is a sub-category of a city, which is a sub-category of Massachusetts).

The query-based search for a respective gazetteer can be performed on the one or more dump files by extracting the sub-classes and/or classes of the named entities included in the content stored in the one or more dump files, determining which sub-classes and/or classes the extracted sub-classes and/or classes correspond to, extracting the sub-classes and/or classes that are similar to the named entity category for the respective gazetteer, and retrieving a set of retrieved named entities by extracting the named entities that are categorized into the extracted sub-classes and/or classes from the one or more dump files. The sub-classes and/or classes can be extracted from the one or more dump files based on the column or columns of the table or tables that includes those sub-classes and/or classes. Additional information for the extracting sub-classes and/or classes from a knowledge base dump file is found in "Self-Attention Gazetteer Embeddings for Named-Entity Recognition" by Peshterlieve et al., published in arXiv preprint arXiv: 2004.04060v2, the entire contents of which are hereby incorporated by reference as if fully set forth herein. In some cases, a sub-class may be too fine grained or granular to be similar to the named entity category for the respective gazetteer. In these cases, the classes to which the extracted sub-classes correspond to can be determined by traversing the Wikidata sub-class/class hierarchy using the publicly available Wikidata-Taxonomy tool. In order to determine whether a sub-class and/or class is similar to the named entity category for the respective gazetteer, a similarity score can be calculated between each of the determined sub-classes and/or classes and the named entity category for the respective gazetteer and any determined sub-classes and/or classes having a similarity score greater than or equal to a predetermined similarity threshold can be determined to be similar to the named entity category and extracted. The similarity score can range between 0 and 1, with a similarity score of 0 indicating that there is no similarity between a determined sub-class and/or class and the named entity category and 1 indicating that a determined sub-class and/or class is identical to the named entity category. The set of retrieved named entities can be retrieved by extracting the named entities that are categorized into the extracted sub-classes and/or classes using a query service such as the Wikidata Query Service. Additional information for query-based searching of publicly available knowledge base dump files is found in "Gazetteer Generation for Neural Named Entity Recognition" by Song et al., published in The Thirty-Third International Flairs Conference, the entire contents of which are hereby incorporated by reference as if fully set forth herein. The query service is configured to extract the named entities and any aliases corresponding to the named entities (e.g., Mass. and Massachusetts) for a given named entity category (e.g., location).

As with the pre-trained model, the set of retrieved named entities retrieved using a query-based search can be processed to produce a set of processed named entities. The set of retrieved named entities can be processed by removing from the set of retrieved named entities artifacts (e.g., parentheticals, punctuation, hyphens, etc.), any named entities that are simply a combination of digits and symbols (i.e., no letters), and any named entities which are duplicative (i.e., either identical to a named entity in a gazetteer or identical to another named entity in the set of retrieved named entities). Additionally, the case of each named entity in the set of retrieved named entities can be converted to lowercase, uppercase, and/or to match the case of the named entities in the gazetteer.

In order to further diversity the augmented training data, the named entities in each respective expanded gazetteer can be perturbed to generate one or more perturbed versions of those named entities. For example, the named entity [Jane Doe] in the person gazetteer ($GZ_{[PER]}$: {[Jane Doe]; . . .) can be perturbed to generate one or more perturbed versions of the named entity (e.g., [Jae Doe], [Jane Do], [Janes Doey], . . .). The one or more perturbed versions of the named entities can be generated with a typographical error generating function. By applying the typographical error generating function to the named entities in a respective expanded gazetteer, one or more perturbed versions of those named entities can be generated and the one or more perturbed versions of those named entity can be added to the respective expanded gazetteer (e.g., ($GZ_{[PER]}$: {[Jane Doe]; [Jae Doe]; [Jane Do]; [Janes Doey]; . . .). An example of a typographical error generating function is included in the NLPaug library. NLPaug is an open-source library that includes tools for textual augmentation. Even though perturbation has been described with respect to a typographical error generating function, this is not intended to be limiting, and other perturbations and perturbation functions can be used.

As discussed above, a gazetteer can be generated for each named entity category of the set of named entity categories. Further, as discussed above, each respective gazetteer can be expanded by searching one or more knowledge bases for named entities that are similar to the named entities in the respective gazetteer using a pre-trained model and inserting the results of the search in the respective gazetteer and further expanded by querying one or more knowledge bases for named entities that are categorized into one or more sub-classes and/or classes that are similar to the named entity category for the respective gazetteer and inserting the results of the query in the respective gazetteer. Additionally, each respective expanded gazetteer can be further diversified by perturbing the named entities in the respective expanded gazetteers using a typographical error generating function. In this way, a gazetteer for each named entity category of the set of named entity categories can include named entities found in the training data but also named entities found in other sources (i.e., sources external to the training data) and named entities that include perturbations. By considering named entities from the training data, sources external to the training data, and perturbed versions of the named entities, NER models trained with training data incorporating these named entities will be able to generalize well to variations of entities and new and unseen entities.

Templates

The process for augmenting the training data continues with using the gazetteers to generate additional utterances which can be added to the training data. The augmented training data includes the original utterances and generated additional utterances. In order to generate additional utterances, the original utterances in the training data can be transformed into template utterances and the template utterances can be populated with named entities selected from the gazetteers that are different from the named entities of the original utterances. An original utterance in the training data can be transformed into a template utterance by removing the named entity or named entities from the original utterance and inserting a placeholder identifier for each removed named entity or named entities. A placeholder identifier can represent the named entity category of the named entity removed for which the placeholder identifier is inserted. For example, the original utterance "My name is [$_{PER}$ Jane Doe]" can be transformed into the template utterance "My name is [$_{PER}$]," where the [$_{PER}$] placeholder identifier for a named entity that corresponds to a person named entity category is inserted into the original utterance because the named entity [Jane Doe] corresponds to a person. A template utterance can be populated with one or more named entities selected from one or more of the gazetteers by replacing each placeholder identifier in the template utterance with a named entity selected from the gazetteer that corresponds the named entity category of the placeholder identifier. For example, continuing with the example above, in order to populate the template utterance "My name is [$_{PER}$]," the placeholder identifier [$_{PER}$] in the template utterance is replaced with a named entity selected from the person gazetteer ($GZ_{[PER]}$: {[Jane Doe]; [Jae Doe]; [Jane Do]; [Janes Doey]; . . .). Once the template utterance is populated, the generated additional utterance is added to the original utterances in the training data to form augmented training data.

A named entity can be selected from a gazetteer randomly and/or using predetermined criteria. In some implementations, predetermined criteria can include selecting a named entity because it is a perturbed version of a named entity in an original utterance, selecting a named entity because there are no original utterances and/or generated additional utterances in the training data with that named entity, and the like. In some implementations, the predetermined criteria can be defined by a user, a customer, a developer, and the like, of a NER model. In some implementations, additional utterances can be generated such that each named entity in each gazetteer is used in at least one utterance of the augmented training data. In other implementations, the number of generated additional utterances can be variable and/or depend upon the number of original utterances in the training data. For example, the number of generated additional utterances can correspond to a predetermined percentage (e.g., 500%) of the number of original utterances in the training data. The augmented training data can be used to train the NER model.

NER Model

As discussed above with respect to FIG. **4**A, the model training subsystem **440** outputs trained models including one or more trained NER models **455** that have been trained with the augmented training data. The one or more trained NER models **455** can be implemented in a system or artificial intelligence-based solution such as a chatbot system. A simplified block diagram of a system **470** in which a trained NER model **472** can be implemented is shown in FIG. **4**B. As shown in FIG. **4**B, the system **470** can include the trained NER model **472**. Using the trained NER model **472**, the system **470** can receive an input utterance **480** (e.g., My name is John and I live in Texas) and generate an output utterance **490** that includes the input utterance **480** labeled with labels that identify the named entities in the input utterance **480** and the categories to which the identified named entities belong (e.g., My name is [$_{PER}$ John] and I live in [$_{LOCATION}$ Texas]).

The trained NER model **472** can be based on a transformer-based model, such as a bidirectional encoder representations from transformers (BERT) model. In some implementations, the trained NER Model **472** can include a BERT layer **474**, a convolutional neural network/bidirectional long short-term memory (CNN/BiLSTM) layer **476**, and a conditional random field (CRF) layer **478**.

The BERT layer **474** can include an algorithm that accepts a sequence of words of an utterance such as the input utterance **480** as an input and generates feature vectors or word embeddings for the words of the sequence. In some implementations, the BERT layer **474** includes a transformer layer that includes encoders (e.g., attention mechanisms and feed-forward networks). The attention mechanisms can generate attention scores for the words of the sequence and the feed-forward networks can encode the words of the sequence into word embeddings based on the attention scores.

The CNN/BILSTM layer **476** can include an algorithm that accepts the word embeddings from the BERT layer **474** and generates a sentence or string embedding for the sequence of words. The CNN portion can accept the word embeddings from the BERT layer **474** and generate character-level vector representations for each character of each word of the words of the sequence and the BiLSTM portion can accept the word embeddings from the BERT layer **474** and the character-level vector representations from the CNN portion and generate a sentence or string embedding for the sequence of words.

The CRF layer **478** can include an algorithm that accepts as an input the sentence or string embedding, detects whether or not a word of the sequence of words is a named entity, and classifies the detected named entities as the type of named entity to which it belongs. In some implementations, the CFR layer **478** can output the classified detected named entities and identifiers that identify which words in the sequence of words that the detected named entities correspond to.

In some implementations, the system **470** can receive the detected named entities and the identifiers from the NER model **472** and generate an output utterance such as the output utterance **490** that is labeled with the named entities contained therein (e.g., My name is [$_{PER}$ John] and I live in [$_{LOCATION}$ Texas]). The system **470** can use the labeled output utterance **490** to perform an operation such as query a database, generate a response, and display, on a display device of the system **470** or a client device (not shown), the generated response. With training data augmentation techniques described herein, the NER model **472** can accurately identify named entities in an input utterance which the enables the system **470** to generate improved responses along with improved performance of other functions (e.g., understanding a language).

The described construction, operations, functions, and advantages of the system **470** and the NER model **472** are not intended to be limiting, and other construction, operations, functions, and advantages are possible and included in this description. Furthermore, while not explicitly shown, it will be appreciated that the system **470** may further include a developer device associated with a developer. Communications from a developer device to components of the system **470** may indicate what types of input data and/or utterances are to be used for the models, a number and type of models to be used, hyperparameters of each model, for example, learning rate and number of hidden layers, how data requests are to be formatted, which training data is to be used (e.g., and how to gain access to the training data) and which validation technique is to be used, and/or how the controller processes are to be configured.

Illustrative Methods

FIG. **5**A is a flowchart that illustrates an example process **500** for augmenting training data according to certain embodiments. The processing depicted in FIG. **5**A may be implemented in software (e.g., code, instructions, program) executed by one or more processing units (e.g., processors, cores) of the respective systems, hardware, or combinations thereof. The software may be stored on a non-transitory storage medium (e.g., on a memory device). The method presented in FIG. **5**A and described below is intended to be illustrative and non-limiting. Although FIG. **5**A depicts the various processing steps occurring in a particular sequence or order, this is not intended to be limiting. In certain alternative embodiments, the steps may be performed in some different order or some steps may also be performed in parallel. In certain embodiments, such as in the embodiment depicted in FIGS. **1**-**4**B, the processing depicted in FIG. **5**A may be performed by a pre-processing subsystem (e.g., pre-processing subsystem **210** or prediction model training subsystem **410**) to generate augmented training data for training an NER model.

At step **502**, training data is obtained. The training data includes original utterances. Each original utterance in the obtained training data can include one or more named entities corresponding to one or more named entity categories of a set of named entity categories. Each named entity in an original utterance can include a label that indicates the category to which the named entity belongs. Each category of the set of named entity categories can be associated with a different label. For example, a person named entity category can be associated with a [PER] label and a location named entity category can be associated with a [LOC]. Each utterance in the training data can also include contextual information before and/or after the one or more named entities in the utterance. For example, for the utterance "His name is [PER John Doe]," the "His name is" portion of the utterance is contextual information that is before the named entity [PER John Doe].

At step **504**, gazetteers are generated. A gazetteer can be generated for a named entity category by extracting the named entities from the utterances of the training data that correspond to the respective named entity category and inserting the extracted named entities into the gazetteer for the respective category. Each respective gazetteer can be expanded by searching one or more sources other than the training data for named entities that are similar to the named entities in the respective gazetteer and inserting the results of the search (i.e., the similar named entities) in the respective gazetteer. An example of a source other than the training data is a knowledge base, which can include any publicly available knowledge base or combination of publicly available knowledge bases (e.g., Wikipedia, Wikidata, Wikimedia, DBpedia).

The knowledge base can be searched by searching one or more publicly available dump files (i.e., record files, archival files, etc.) for the knowledge base using a pre-trained model. The pre-trained model can be configured to receive a gazetteer and one or more dump files for a knowledge base or knowledge bases to be searched as inputs, retrieve a set of retrieved named entities by searching the one or more dump files for named entities that are similar to the named entities in the gazetteer, processing the set of retrieved named entities to produce a set of processed named entities, and combining the set of processed named entities with the named entities in the gazetteer. The pre-trained model can be configured to calculate a set of similarity scores for the set of retrieved named entities such that a similarity score is calculated for each retrieved named entity in the set of retrieved named entities. The calculated score for a retrieved named entity can represent how similar the retrieved named entity is to any of the named entities in the gazetteer received as an input to pre-trained model. The calculated score can range between 0 and 1, with 0 indicating that there is a no similarity between the retrieved named entity and a named entity in the gazetteer and 1 indicating that the retrieved named entity is identical to a named entity in the gazetteer. In some implementations, retrieved named entities having a similarity greater than or equal to a predetermined similarity threshold can be included in the set of the retrieved named entities. The pre-trained model can be configured to process the set of retrieved named entities to produce a set of processed named entities by removing from the set of retrieved named entities artifacts (e.g., parentheticals, punctuation, hyphens, etc.), any named entities that are a combination of digits and symbols (i.e., no letters), and any named entities that are duplicative (i.e., either identical to a named entity in a gazetteer or identical to another named entity in the set of retrieved named entities). In some implementations, the pre-trained model can be configured to convert the case of each named entity in the set of retrieved named entities to lowercase, uppercase, and/or to match the case of the named entities in the gazetteer.

Additionally, or alternatively, each respective gazetteer can be further expanded by querying one or more knowledge bases for named entities that are categorized into one or more classes that are similar to the named entity category for the respective gazetteer. A knowledge base can be queried by querying the one or more publicly available dump files for the knowledge base using a query-based search. In order to further expand a respective gazetteer, a query-based search can be performed on the one or more dump files for a knowledge base or knowledge bases to be queried to retrieve a set of retrieved named entities that are categorized into one or more sub-classes and/or classes that are similar to the named entity category of the respective gazetteer. The query-based search for a respective gazetteer can be performed on the one or more dump files by extracting the sub-classes and/or classes of the named entities included in the content stored in the one or more dump files, determining which sub-classes and/or classes the extracted sub-classes and/or classes correspond to, extracting the sub-classes and/or classes that are similar to the named entity category for the respective gazetteer, and retrieving a set of retrieved named entities by extracting the named entities that are categorized into the extracted sub-classes and/or classes from the one or more dump files. The sub-classes and/or classes can be extracted from the one or more dump files based on the column or columns of the table or tables that includes those sub-classes and/or classes. In some cases, a sub-class may be too fine grained or granular to be similar to the named entity category for the respective gazetteer. In these cases, the classes to which the extracted sub-classes correspond to can be determined by traversing the knowledge base sub-class/class hierarchy using the publicly available taxonomy tool. In order to determine whether a sub-class and/or class is similar to the named entity category for the respective gazetteer, a similarity score can be calculated between each of the determined sub-classes and/or classes and the named entity category for the respective gazetteer and any determined sub-classes and/or classes having a similarity score greater than or equal to a predetermined similarity threshold can be determined to be similar to the named entity category and extracted. The similarity score can range between 0 and 1, with a similarity score of 0 indicating that there is no similarity between a determined sub-class and/or class and the named entity category and 1 indicating that a determined sub-class and/or class is identical to the named entity category. The set of retrieved named entities can be retrieved by extracting the named entities that are categorized into the extracted sub-classes and/or classes using a query service. The query service is configured to extract the named entities and any aliases corresponding to the named entities (e.g., Mass. and Massachusetts) for a given named entity category (e.g., location). The set of retrieved named entities retrieved using a query-based search can be processed to produce a set of processed named entities. The set of retrieved named entities can be processed by removing from the set of retrieved named entities artifacts (e.g., parentheticals, punctuation, hyphens, etc.), any named entities that are simply a combination of digits and symbols (i.e., no letters), and any named entities which are duplicative (i.e., either identical to a named entity in a gazetteer or identical to another named entity in the set of retrieved named entities). Additionally, the case of each named entity in the set of retrieved named entities can be converted to lowercase, uppercase, and/or to match the case of the named entities in the gazetteer.

In order to further diversity the augmented training data, the named entities in each respective expanded gazetteer can be perturbed to generate one or more perturbed versions of those named entities. The one or more perturbed versions of the named entities can be generated with a typographical error generating function. By applying the typographical error generating function to the named entities in a respective expanded gazetteer, one or more perturbed versions of those named entities can be generated and the one or more perturbed versions of those named entity can be added to the respective expanded gazetteer. Even though perturbation has been described with respect to a typographical error generating function, this is not intended to be limiting, and other perturbations can be used.

At step **506**, template utterances are generated. Each template utterance includes an original utterance of the training data and at least one placeholder identifier corresponding to a named entity in the original utterance. An original utterance in the training data can be transformed into a template utterance by removing the named entity or named entities from the original utterance and inserting a placeholder identifier for each removed named entity or named entities. A placeholder identifier can represent the named entity category of the named entity removed for which the placeholder identifier is inserted.

At step **508**, additional utterances are generated. Each additional utterance includes a template utterance populated with at least one named entity selected from the gazetteers. A template utterance can be populated with one or more named entities selected from one or more of the gazetteers by replacing each placeholder identifier in the template utterance with a named entity selected from the gazetteer that corresponds the named entity category of the placeholder identifier. A named entity can be selected from a gazetteer randomly and/or using predetermined criteria. In some implementations, predetermined criteria can include selecting a named entity because it is a perturbed version of a named entity in an original utterance, selecting a named entity because there are no original utterances and/or generated additional utterances in the training data with that named entity, and the like. In some implementations, the predetermined criteria can be defined by a user, a customer, a developer, and the like, of a NER model. In some implementations, additional utterances can be generated such that each named entity in each gazetteer is used in at least one utterance of the augmented training data. In other implementations, the number of generated additional utterances can be variable and/or depend upon the number of original utterances in the training data. For example, the number of generated additional utterances can correspond to a predetermined percentage (e.g., 500%) of the number of original utterances in the training data.

At step **510**, the training data is augmented. The training data is augmented by adding the additional utterances to the original utterances.

FIG. **5**B is a flowchart that illustrates an example process **550** for providing an NER model according to certain embodiments. The processing depicted in FIG. **5**B may be implemented in software (e.g., code, instructions, program) executed by one or more processing units (e.g., processors, cores) of the respective systems, hardware, or combinations thereof. The software may be stored on a non-transitory storage medium (e.g., on a memory device). The method presented in FIG. **5**B and described below is intended to be illustrative and non-limiting. Although FIG. **5**B depicts the various processing steps occurring in a particular sequence or order, this is not intended to be limiting. In certain alternative embodiments, the steps may be performed in some different order or some steps may also be performed in parallel. In certain embodiments, such as in the embodiment depicted in FIGS. **1**-**4**B, the processing depicted in FIG. **5**B may be performed by a processing subsystem (e.g., prediction model training **410**, evaluation subsystem **415**, and/or implementation subsystem **420**) to train an NER model with augmented training data and deploy the trained NER model.

At step **552**, an NER model is trained. The NER model can be trained with training data obtained and/or augmented described above. Training the NER model includes selecting hyperparameters for the NER model and performing iterative operations of inputting utterances from the augmented training data into the NER model to find a set of model parameters (e.g., weights and/or biases) that maximizes or minimizes an objective function, e.g., minimizes a loss function, for the NER model. Each iteration of training can involve finding a set of model parameters for the NER model (configured with a defined set of hyperparameters) so that the value of the objective function using the set of model parameters is smaller than the value of the objective function using a different set of model parameters in a previous iteration. The objective function can be constructed to measure the difference between the outputs inferred using the NER model and the augmented training data using the labels. Once the set of model parameters are identified, the NER model has been trained and can be tested or validated using the subset of testing data (testing or validation data set). The testing or validation process includes iterative operations of inputting utterances from the subset of testing data into the NER model using a validation technique such as K-Fold Cross-Validation, Leave-one-out Cross-Validation, Leave-one-group-out Cross-Validation, Nested Cross-Validation, or the like to tune the hyperparameters and ultimately find the optimal set of hyperparameters. Once the optimal set of hyperparameters are obtained, a reserved test set from the subset of test data may be input into the NER model to obtain output (in this example, one or more recognized entities), and the output is evaluated versus ground truth entities using correlation techniques such as Bland-Altman method and the Spearman's rank correlation coefficients. Further, performance metrics may be calculated such as the error, accuracy, precision, recall, receiver operating characteristic curve (ROC), etc. The performance metrics may be used to analyze performance of the NER model.

At step **554**, the trained NER model is provided. The NER model can be provided to or deployed or implemented in a system or artificial intelligence-based solution such as a chatbot system. Using the trained NER model, the system can receive an input utterance and generate an output utterance that includes the input utterance labeled with labels that identify the named entities in the input utterance and the categories to which the identified named entities belong. The system can use the labeled output utterance to perform an operation such as query a database, generate a response, and display, on a display device of the system or a client device (not shown), the generated response.

Illustrative Systems

FIG. **6** depicts a simplified diagram of a distributed system **600**. In the illustrated example, distributed system **600** includes one or more client computing devices **602**, **604**, **606**, and **608**, coupled to a server **612** via one or more communication networks **610**. Clients computing devices **602**, **604**, **606**, and **608** may be configured to execute one or more applications.

In various examples, server **612** may be adapted to run one or more services or software applications that enable one or more embodiments described in this disclosure. In certain examples, server **612** may also provide other services or software applications that may include non-virtual and virtual environments. In some examples, these services may be offered as web-based or cloud services, such as under a Software as a Service (SaaS) model to the users of client computing devices **602**, **604**, **606**, and/or **608**. Users operating client computing devices **602**, **604**, **606**, and/or **608** may in turn utilize one or more client applications to interact with server **612** to utilize the services provided by these components.

In the configuration depicted in FIG. **6**, server **612** may include one or more components **618**, **620** and **622** that implement the functions performed by server **612**. These components may include software components that may be executed by one or more processors, hardware components, or combinations thereof. It should be appreciated that various different system configurations are possible, which may be different from distributed system **600**. The example shown in FIG. **6** is thus one example of a distributed system for implementing an example system and is not intended to be limiting.

Users may use client computing devices **602**, **604**, **606**, and/or **608** to execute one or more applications, models or chatbots, which may generate one or more events or models that may then be implemented or serviced in accordance with the teachings of this disclosure. A client device may provide an interface that enables a user of the client device to interact with the client device. The client device may also output information to the user via this interface. Although FIG. **6** depicts only four client computing devices, any number of client computing devices may be supported.

The client devices may include various types of computing systems such as portable handheld devices, general purpose computers such as personal computers and laptops, workstation computers, wearable devices, gaming systems, thin clients, various messaging devices, sensors or other sensing devices, and the like. These computing devices may run various types and versions of software applications and operating systems (e.g., Microsoft Windows®, Apple Macintosh®, UNIX® or UNIX-like operating systems, Linux or Linux-like operating systems such as Google Chrome™ OS) including various mobile operating systems (e.g., Microsoft Windows Mobile®, iOS®, Windows Phone®, Android™, BlackBerry®, Palm OS®). Portable handheld devices may include cellular phones, smartphones, (e.g., an iPhone®), tablets (e.g., iPad®), personal digital assistants (PDAs), and the like. Wearable devices may include Google Glass® head mounted display, and other devices. Gaming systems may include various handheld gaming devices, Internet-enabled gaming devices (e.g., a Microsoft Xbox® gaming console with or without a Kinect® gesture input device, Sony PlayStation® system, various gaming systems provided by Nintendo®, and others), and the like. The client devices may be capable of executing various different applications such as various Internet-related apps, communication applications (e.g., E-mail applications, short message service (SMS) applications) and may use various communication protocols.

Network(s) **610** may be any type of network familiar to those skilled in the art that may support data communications using any of a variety of available protocols, including without limitation TCP/IP (transmission control protocol/Internet protocol), SNA (systems network architecture), IPX (Internet packet exchange), AppleTalk®, and the like. Merely by way of example, network(s) **610** may be a local area network (LAN), networks based on Ethernet, Token-Ring, a wide-area network (WAN), the Internet, a virtual network, a virtual private network (VPN), an intranet, an extranet, a public switched telephone network (PSTN), an infra-red network, a wireless network (e.g., a network operating under any of the Institute of Electrical and Electronics (IEEE) 1002.11 suite of protocols, Bluetooth®, and/or any other wireless protocol), and/or any combination of these and/or other networks.

Server **612** may be composed of one or more general purpose computers, specialized server computers (including, by way of example, PC (personal computer) servers, UNIX® servers, mid-range servers, mainframe computers, rack-mounted servers, etc.), server farms, server clusters, or any other appropriate arrangement and/or combination. Server **612** may include one or more virtual machines running virtual operating systems, or other computing architectures involving virtualization such as one or more flexible pools of logical storage devices that may be virtualized to maintain virtual storage devices for the server. In various examples, server **612** may be adapted to run one or more services or software applications that provide the functionality described in the foregoing disclosure.

The computing systems in server **612** may run one or more operating systems including any of those discussed above, as well as any commercially available server operating system. Server **612** may also run any of a variety of additional server applications and/or mid-tier applications, including HTTP (hypertext transport protocol) servers, FTP (file transfer protocol) servers, CGI (common gateway interface) servers, JAVA® servers, database servers, and the like. Exemplary database servers include without limitation those commercially available from Oracle®, Microsoft®, Sybase®, IBM® (International Business Machines), and the like.

In some implementations, server **612** may include one or more applications to analyze and consolidate data feeds and/or event updates received from users of client computing devices **602**, **604**, **606**, and **608**. As an example, data feeds and/or event updates may include, but are not limited to, Twitter® feeds, Facebook® updates or real-time updates received from one or more third party information sources and continuous data streams, which may include real-time events related to sensor data applications, financial tickers, network performance measuring tools (e.g., network monitoring and traffic management applications), clickstream analysis tools, automobile traffic monitoring, and the like. Server **612** may also include one or more applications to display the data feeds and/or real-time events via one or more display devices of client computing devices **602**, **604**, **606**, and **608**.

Distributed system **600** may also include one or more data repositories **614**, **616**. These data repositories may be used to store data and other information in certain examples. For example, one or more of the data repositories **614**, **616** may be used to store information such as information related to chatbot performance or generated models for use by chatbots used by server **612** when performing various functions in accordance with various embodiments. Data repositories **614**, **616** may reside in a variety of locations. For example, a data repository used by server **612** may be local to server **612** or may be remote from server **612** and in communication with server **612** via a network-based or dedicated connection. Data repositories **614**, **616** may be of different types. In certain examples, a data repository used by server **612** may be a database, for example, a relational database, such as databases provided by Oracle Corporation® and other vendors. One or more of these databases may be adapted to enable storage, update, and retrieval of data to and from the database in response to SQL-formatted commands.

In certain examples, one or more of data repositories **614**, **616** may also be used by applications to store application data. The data repositories used by applications may be of different types such as, for example, a key-value store repository, an object store repository, or a general storage repository supported by a file system.

In certain examples, the functionalities described in this disclosure may be offered as services via a cloud environment. FIG. **7** is a simplified block diagram of a cloud-based system environment in which various services may be offered as cloud services in accordance with certain examples. In the example depicted in FIG. **7**, cloud infrastructure system **702** may provide one or more cloud services that may be requested by users using one or more client computing devices **704**, **706**, and **708**. Cloud infrastructure system **702** may comprise one or more computers and/or servers that may include those described above for server **612**. The computers in cloud infrastructure system **702** may be organized as general-purpose computers, specialized server computers, server farms, server clusters, or any other appropriate arrangement and/or combination.

Network(s) **710** may facilitate communication and exchange of data between clients **704**, **706**, and **708** and cloud infrastructure system **702**. Network(s) **710** may include one or more networks. The networks may be of the same or different types. Network(s) **710** may support one or more communication protocols, including wired and/or wireless protocols, for facilitating the communications.

The example depicted in FIG. **7** is only one example of a cloud infrastructure system and is not intended to be limiting. It should be appreciated that, in some other examples, cloud infrastructure system **702** may have more or fewer components than those depicted in FIG. **7**, may combine two or more components, or may have a different configuration or arrangement of components. For example, although FIG. **7** depicts three client computing devices, any number of client computing devices may be supported in alternative examples.

The term cloud service is generally used to refer to a service that is made available to users on demand and via a communication network such as the Internet by systems (e.g., cloud infrastructure system **702**) of a service provider. Typically, in a public cloud environment, servers and systems that make up the cloud service provider's system are different from the customer's own on-premise servers and systems. The cloud service provider's systems are managed by the cloud service provider. Customers may thus avail themselves of cloud services provided by a cloud service provider without having to purchase separate licenses, support, or hardware and software resources for the services. For example, a cloud service provider's system may host an application, and a user may, via the Internet, on demand, order and use the application without the user having to buy infrastructure resources for executing the application. Cloud services are designed to provide easy, scalable access to applications, resources, and services. Several providers offer cloud services. For example, several cloud services are offered by Oracle Corporation® of Redwood Shores, California, such as middleware services, database services, Java cloud services, and others.

In certain examples, cloud infrastructure system **702** may provide one or more cloud services using different models such as under a Software as a Service (SaaS) model, a Platform as a Service (PaaS) model, an Infrastructure as a Service (IaaS) model, and others, including hybrid service models. Cloud infrastructure system **702** may include a suite of applications, middleware, databases, and other resources that enable provision of the various cloud services.

A SaaS model enables an application or software to be delivered to a customer over a communication network like the Internet, as a service, without the customer having to buy the hardware or software for the underlying application. For example, a SaaS model may be used to provide customers access to on-demand applications that are hosted by cloud infrastructure system **702**. Examples of SaaS services provided by Oracle Corporation® include, without limitation, various services for human resources/capital management, customer relationship management (CRM), enterprise resource planning (ERP), supply chain management (SCM), enterprise performance management (EPM), analytics services, social applications, and others.

An IaaS model is generally used to provide infrastructure resources (e.g., servers, storage, hardware and networking resources) to a customer as a cloud service to provide elastic compute and storage capabilities. Various IaaS services are provided by Oracle Corporation®.

A PaaS model is generally used to provide, as a service, platform and environment resources that enable customers to develop, run, and manage applications and services without the customer having to procure, build, or maintain such resources. Examples of PaaS services provided by Oracle Corporation® include, without limitation, Oracle Java Cloud Service (JCS), Oracle Database Cloud Service (DBCS), data management cloud service, various application development solutions services, and others.

Cloud services are generally provided on an on-demand self-service basis, subscription-based, elastically scalable, reliable, highly available, and secure manner. For example, a customer, via a subscription order, may order one or more services provided by cloud infrastructure system **702**. Cloud infrastructure system **702** then performs processing to provide the services requested in the customer's subscription order. For example, a user may use utterances to request the cloud infrastructure system to take a certain action (e.g., an intent), as described above, and/or provide services for a chatbot system as described herein. Cloud infrastructure system **702** may be configured to provide one or even multiple cloud services.

Cloud infrastructure system **702** may provide the cloud services via different deployment models. In a public cloud model, cloud infrastructure system **702** may be owned by a third party cloud services provider and the cloud services are offered to any general public customer, where the customer may be an individual or an enterprise. In certain other examples, under a private cloud model, cloud infrastructure system **702** may be operated within an organization (e.g., within an enterprise organization) and services provided to customers that are within the organization. For example, the customers may be various departments of an enterprise such as the Human Resources department, the Payroll department, etc. or even individuals within the enterprise. In certain other examples, under a community cloud model, the cloud infrastructure system **702** and the services provided may be shared by several organizations in a related community. Various other models such as hybrids of the above mentioned models may also be used.

Client computing devices **704**, **706**, and **708** may be of different types (such as client computing devices **602**, **604**, **606**, and **608** depicted in FIG. **6**) and may be capable of operating one or more client applications. A user may use a client device to interact with cloud infrastructure system **702**, such as to request a service provided by cloud infrastructure system **702**. For example, a user may use a client device to request information or action from a chatbot as described in this disclosure.

In some examples, the processing performed by cloud infrastructure system **702** for providing services may involve model training and deployment. This analysis may involve using, analyzing, and manipulating data sets to train and deploy one or more models. This analysis may be performed by one or more processors, possibly processing the data in parallel, performing simulations using the data, and the like. For example, big data analysis may be performed by cloud infrastructure system **702** for generating and training one or more models for a chatbot system. The data used for this analysis may include structured data (e.g., data stored in a database or structured according to a structured model) and/or unstructured data (e.g., data blobs (binary large objects)).

As depicted in the example in FIG. 7, cloud infrastructure system **702** may include infrastructure resources **730** that are utilized for facilitating the provision of various cloud services offered by cloud infrastructure system **702**. Infrastructure resources **730** may include, for example, processing resources, storage or memory resources, networking resources, and the like. In certain examples, the storage virtual machines that are available for servicing storage requested from applications may be part of cloud infrastructure system **702**. In other examples, the storage virtual machines may be part of different systems.

In certain examples, to facilitate efficient provisioning of these resources for supporting the various cloud services provided by cloud infrastructure system **702** for different customers, the resources may be bundled into sets of resources or resource modules (also referred to as "pods"). Each resource module or pod may comprise a pre-integrated and optimized combination of resources of one or more types. In certain examples, different pods may be pre-provisioned for different types of cloud services. For example, a first set of pods may be provisioned for a database service, a second set of pods, which may include a different combination of resources than a pod in the first set of pods, may be provisioned for Java service, and the like. For some services, the resources allocated for provisioning the services may be shared between the services.

Cloud infrastructure system **702** may itself internally use services **732** that are shared by different components of cloud infrastructure system **702** and which facilitate the provisioning of services by cloud infrastructure system **702**. These internal shared services may include, without limitation, a security and identity service, an integration service, an enterprise repository service, an enterprise manager service, a virus scanning and whitelist service, a high availability, backup and recovery service, service for enabling cloud support, an email service, a notification service, a file transfer service, and the like.

Cloud infrastructure system **702** may comprise multiple subsystems. These subsystems may be implemented in software, or hardware, or combinations thereof. As depicted in FIG. **7**, the subsystems may include a user interface subsystem **712** that enables users or customers of cloud infrastructure system **702** to interact with cloud infrastructure system **702**. User interface subsystem **712** may include various different interfaces such as a web interface **714**, an online store interface **716** where cloud services provided by cloud infrastructure system **702** are advertised and are purchasable by a consumer, and other interfaces **718**. For example, a customer may, using a client device, request (service request **734**) one or more services provided by cloud infrastructure system **702** using one or more of interfaces **714**, **716**, and **718**. For example, a customer may access the online store, browse cloud services offered by cloud infrastructure system **702**, and place a subscription order for one or more services offered by cloud infrastructure system **702** that the customer wishes to subscribe to. The service request may include information identifying the customer and one or more services that the customer desires to subscribe to. For example, a customer may place a subscription order for a service offered by cloud infrastructure system **702**. As part of the order, the customer may provide information identifying a chatbot system for which the service is to be provided and optionally one or more credentials for the chatbot system.

In certain examples, such as the example depicted in FIG. **7**, cloud infrastructure system **702** may comprise an order management subsystem (OMS) **720** that is configured to process the new order. As part of this processing, OMS **720** may be configured to: create an account for the customer, if not done already; receive billing and/or accounting information from the customer that is to be used for billing the customer for providing the requested service to the customer; verify the customer information; upon verification, book the order for the customer; and orchestrate various workflows to prepare the order for provisioning.

Once properly validated, OMS **720** may then invoke the order provisioning subsystem (OPS) **724** that is configured to provision resources for the order including processing, memory, and networking resources. The provisioning may include allocating resources for the order and configuring the resources to facilitate the service requested by the customer order. The manner in which resources are provisioned for an order and the type of the provisioned resources may depend upon the type of cloud service that has been ordered by the customer. For example, according to one workflow, OPS **724** may be configured to determine the particular cloud service being requested and identify a number of pods that may have been pre-configured for that particular cloud service. The number of pods that are allocated for an order may depend upon the size/amount/level/scope of the requested service. For example, the number of pods to be allocated may be determined based upon the number of users to be supported by the service, the duration of time for which the service is being requested, and the like. The allocated pods may then be customized for the particular requesting customer for providing the requested service.

In certain examples, setup phase processing, as described above, may be performed by cloud infrastructure system **702** as part of the provisioning process. Cloud infrastructure system **702** may generate an application ID and select a storage virtual machine for an application from among storage virtual machines provided by cloud infrastructure system **702** itself or from storage virtual machines provided by other systems other than cloud infrastructure system **702**.

Cloud infrastructure system **702** may send a response or notification **744** to the requesting customer to indicate when the requested service is now ready for use. In some instances, information (e.g., a link) may be sent to the customer that enables the customer to start using and availing the benefits of the requested services. In certain examples, for a customer requesting the service, the response may include a chatbot system ID generated by cloud infrastructure system **702** and information identifying a chatbot system selected by cloud infrastructure system **702** for the chatbot system corresponding to the chatbot system ID.

Cloud infrastructure system **702** may provide services to multiple customers. For each customer, cloud infrastructure system **702** is responsible for managing information related to one or more subscription orders received from the customer, maintaining customer data related to the orders, and providing the requested services to the customer. Cloud infrastructure system **702** may also collect usage statistics regarding a customer's use of subscribed services. For example, statistics may be collected for the amount of storage used, the amount of data transferred, the number of users, and the amount of system up time and system down time, and the like. This usage information may be used to bill the customer. Billing may be done, for example, on a monthly cycle.

Cloud infrastructure system **702** may provide services to multiple customers in parallel. Cloud infrastructure system **702** may store information for these customers, including possibly proprietary information. In certain examples, cloud infrastructure system **702** comprises an identity management subsystem (IMS) **728** that is configured to manage customer information and provide the separation of the managed information such that information related to one customer is not accessible by another customer. IMS **728** may be configured to provide various security-related services such as identity services, such as information access management, authentication and authorization services, services for managing customer identities and roles and related capabilities, and the like.

FIG. **8** illustrates an example of computer system **800**. In some examples, computer system **800** may be used to implement any of the digital assistant or chatbot systems within a distributed environment, and various servers and computer systems described above. As shown in FIG. **8**, computer system **800** includes various subsystems including a processing subsystem **804** that communicates with a number of other subsystems via a bus subsystem **802**. These other subsystems may include a processing acceleration unit **806**, and I/O subsystem **808**, a storage subsystem **818**, and a communications subsystem **824**. Storage subsystem **818** may include non-transitory computer-readable storage media including storage media **822** and a system memory **810**.

Bus subsystem **802** provides a mechanism for letting the various components and subsystems of computer system **800** communicate with each other as intended. Although bus subsystem **802** is shown schematically as a single bus, alternative examples of the bus subsystem may utilize multiple buses. Bus subsystem **802** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, a local bus using any of a variety of bus architectures, and the like. For example, such architectures may include an Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus, which may be implemented as a Mezzanine bus manufactured to the IEEE P1386.1 standard, and the like.

Processing subsystem **804** controls the operation of computer system **800** and may comprise one or more processors, application specific integrated circuits (ASICs), or field programmable gate arrays (FPGAs). The processors may include be single core or multicore processors. The processing resources of computer system **800** may be organized into one or more processing units **832**, **834**, etc. A processing unit may include one or more processors, one or more cores from the same or different processors, a combination of cores and processors, or other combinations of cores and processors. In some examples, processing subsystem **804** may include one or more special purpose co-processors such as graphics processors, digital signal processors (DSPs), or the like. In some examples, some or all of the processing units of processing subsystem **804** may be implemented using customized circuits, such as application specific integrated circuits (ASICs), or field programmable gate arrays (FPGAs).

In some examples, the processing units in processing subsystem **804** may execute instructions stored in system memory **810** or on computer readable storage media **822**. In various examples, the processing units may execute a variety of programs or code instructions and may maintain multiple concurrently executing programs or processes. At any given time, some or all of the program code to be executed may be resident in system memory **810** and/or on computer-readable storage media **822** including potentially on one or more storage devices. Through suitable programming, processing subsystem **804** may provide various functionalities described above. In instances where computer system **800** is executing one or more virtual machines, one or more processing units may be allocated to each virtual machine.

In certain examples, a processing acceleration unit **806** may optionally be provided for performing customized processing or for off-loading some of the processing performed by processing subsystem **804** so as to accelerate the overall processing performed by computer system **800**.

I/O subsystem **808** may include devices and mechanisms for inputting information to computer system **800** and/or for outputting information from or via computer system **800**. In general, use of the term input device is intended to include all possible types of devices and mechanisms for inputting information to computer system **800**. User interface input devices may include, for example, a keyboard, pointing devices such as a mouse or trackball, a touchpad or touch screen incorporated into a display, a scroll wheel, a click wheel, a dial, a button, a switch, a keypad, audio input devices with voice command recognition systems, microphones, and other types of input devices. User interface input devices may also include motion sensing and/or gesture recognition devices such as the Microsoft Kinect® motion sensor that enables users to control and interact with an input device, the Microsoft Xbox® 360 game controller, devices that provide an interface for receiving input using gestures and spoken commands. User interface input devices may also include eye gesture recognition devices such as the Google Glass® blink detector that detects eye activity (e.g., "blinking" while taking pictures and/or making a menu selection) from users and transforms the eye gestures as inputs to an input device (e.g., Google Glass®). Additionally, user interface input devices may include voice recognition sensing devices that enable users to interact with voice recognition systems (e.g., Siri® navigator) through voice commands.

Other examples of user interface input devices include, without limitation, three dimensional (3D) mice, joysticks or pointing sticks, gamepads and graphic tablets, and audio/visual devices such as speakers, digital cameras, digital camcorders, portable media players, webcams, image scanners, fingerprint scanners, barcode reader 3D scanners, 3D printers, laser rangefinders, and eye gaze tracking devices. Additionally, user interface input devices may include, for example, medical imaging input devices such as computed tomography, magnetic resonance imaging, position emission tomography, and medical ultrasonography devices. User interface input devices may also include, for example, audio input devices such as MIDI keyboards, digital musical instruments and the like.

In general, use of the term output device is intended to include all possible types of devices and mechanisms for outputting information from computer system **800** to a user or other computer. User interface output devices may include a display subsystem, indicator lights, or non-visual displays such as audio output devices, etc. The display subsystem may be a cathode ray tube (CRT), a flat-panel device, such as that using a liquid crystal display (LCD) or plasma display, a projection device, a touch screen, and the like. For example, user interface output devices may include, without limitation, a variety of display devices that visually convey text, graphics, and audio/video information such as monitors, printers, speakers, headphones, automotive navigation systems, plotters, voice output devices, and modems.

Storage subsystem **818** provides a repository or data store for storing information and data that is used by computer system **800**. Storage subsystem **818** provides a tangible non-transitory computer-readable storage medium for storing the basic programming and data constructs that provide the functionality of some examples. Storage subsystem **818** may store software (e.g., programs, code modules, instructions) that when executed by processing subsystem **804** provides the functionality described above. The software may be executed by one or more processing units of processing subsystem **804**. Storage subsystem **818** may also provide authentication in accordance with the teachings of this disclosure.

Storage subsystem **818** may include one or more non-transitory memory devices, including volatile and non-volatile memory devices. As shown in FIG. **8**, storage subsystem **818** includes a system memory **810** and a computer-readable storage media **822**. System memory **810** may include a number of memories including a volatile main random access memory (RAM) for storage of instructions and data during program execution and a non-volatile read only memory (ROM) or flash memory in which fixed instructions are stored. In some implementations, a basic input/output system (BIOS), containing the basic routines that help to transfer information between elements within computer system **800**, such as during start-up, may typically be stored in the ROM. The RAM typically contains data and/or program modules that are presently being operated and executed by processing subsystem **804**. In some implementations, system memory **810** may include multiple different types of memory, such as static random access memory (SRAM), dynamic random access memory (DRAM), and the like.

By way of example, and not limitation, as depicted in FIG. **8**, system memory **810** may load application programs **812** that are being executed, which may include various applications such as Web browsers, mid-tier applications, relational database management systems (RDBMS), etc., program data **814**, and an operating system **816**. By way of example, operating system **816** may include various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems, a variety of commercially-available UNIX® or UNIX-like operating systems (including without limitation the variety of GNU/Linux operating systems, the Google Chrome® OS, and the like) and/or mobile operating systems such as iOS, Windows® Phone, Android® OS, BlackBerry® OS, Palm® OS operating systems, and others.

Computer-readable storage media **822** may store programming and data constructs that provide the functionality of some examples. Computer-readable media **822** may provide storage of computer-readable instructions, data structures, program modules, and other data for computer system **800**. Software (programs, code modules, instructions) that, when executed by processing subsystem **804** provides the functionality described above, may be stored in storage subsystem **818**. By way of example, computer-readable storage media **822** may include non-volatile memory such as a hard disk drive, a magnetic disk drive, an optical disk drive such as a CD ROM, DVD, a Blu-Ray® disk, or other optical media. Computer-readable storage media **822** may include, but is not limited to, Zip® drives, flash memory cards, universal serial bus (USB) flash drives, secure digital (SD) cards, DVD disks, digital video tape, and the like. Computer-readable storage media **822** may also include, solid-state drives (SSD) based on non-volatile memory such as flash-memory based SSDs, enterprise flash drives, solid state ROM, and the like, SSDs based on volatile memory such as solid state RAM, dynamic RAM, static RAM, DRAM-based SSDs, magneto resistive RAM (MRAM) SSDs, and hybrid SSDs that use a combination of DRAM and flash memory based SSDs.

In certain examples, storage subsystem **818** may also include a computer-readable storage media reader **820** that may further be connected to computer-readable storage media **822**. Reader **820** may receive and be configured to read data from a memory device such as a disk, a flash drive, etc.

In certain examples, computer system **800** may support virtualization technologies, including but not limited to virtualization of processing and memory resources. For example, computer system **800** may provide support for executing one or more virtual machines. In certain examples, computer system **800** may execute a program such as a hypervisor that facilitated the configuring and managing of the virtual machines. Each virtual machine may be allocated memory, compute (e.g., processors, cores), I/O, and networking resources. Each virtual machine generally runs independently of the other virtual machines. A virtual machine typically runs its own operating system, which may be the same as or different from the operating systems executed by other virtual machines executed by computer system **800**. Accordingly, multiple operating systems may potentially be run concurrently by computer system **800**.

Communications subsystem **824** provides an interface to other computer systems and networks. Communications subsystem **824** serves as an interface for receiving data from and transmitting data to other systems from computer system **800**. For example, communications subsystem **824** may enable computer system **800** to establish a communication channel to one or more client devices via the Internet for receiving and sending information from and to the client devices. For example, when computer system **800** is used to implement bot system **120** depicted in FIG. **1**, the communication subsystem may be used to communicate with a chatbot system selected for an application.

Communication subsystem **824** may support both wired and/or wireless communication protocols. In certain examples, communications subsystem **824** may include radio frequency (RF) transceiver components for accessing wireless voice and/or data networks (e.g., using cellular telephone technology, advanced data network technology, such as 3G, 4G or EDGE (enhanced data rates for global evolution), Wi-Fi (IEEE 802.XX family standards, or other mobile communication technologies, or any combination thereof), global positioning system (GPS) receiver components, and/or other components. In some examples, communications subsystem **824** may provide wired network connectivity (e.g., Ethernet) in addition to or instead of a wireless interface.

Communication subsystem **824** may receive and transmit data in various forms. In some examples, in addition to other forms, communications subsystem **824** may receive input communications in the form of structured and/or unstructured data feeds **826**, event streams **828**, event updates **830**, and the like. For example, communications subsystem **824** may be configured to receive (or send) data feeds **826** in real-time from users of social media networks and/or other communication services such as Twitter® feeds, Facebook®

updates, web feeds such as Rich Site Summary (RSS) feeds, and/or real-time updates from one or more third party information sources.

In certain examples, communications subsystem **824** may be configured to receive data in the form of continuous data streams, which may include event streams **828** of real-time events and/or event updates **830**, that may be continuous or unbounded in nature with no explicit end. Examples of applications that generate continuous data may include, for example, sensor data applications, financial tickers, network performance measuring tools (e.g., network monitoring and traffic management applications), clickstream analysis tools, automobile traffic monitoring, and the like.

Communications subsystem **824** may also be configured to communicate data from computer system **800** to other computer systems or networks. The data may be communicated in various different forms such as structured and/or unstructured data feeds **826**, event streams **828**, event updates **830**, and the like to one or more databases that may be in communication with one or more streaming data source computers coupled to computer system **800**.

Computer system **800** may be one of various types, including a handheld portable device (e.g., an iPhone® cellular phone, an iPad® computing tablet, a PDA), a wearable device (e.g., a Google Glass® head mounted display), a personal computer, a workstation, a mainframe, a kiosk, a server rack, or any other data processing system. Due to the ever-changing nature of computers and networks, the description of computer system **800** depicted in FIG. **8** is intended only as a specific example. Many other configurations having more or fewer components than the system depicted in FIG. **8** are possible. Based on the disclosure and teachings provided herein, it should be appreciated there are other ways and/or methods to implement the various examples.

Although specific examples have been described, various modifications, alterations, alternative constructions, and equivalents are possible. Examples are not restricted to operation within certain specific data processing environments but are free to operate within a plurality of data processing environments. Additionally, although certain examples have been described using a particular series of transactions and steps, it should be apparent to those skilled in the art that this is not intended to be limiting. Although some flowcharts describe operations as a sequential process, many of the operations may be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure. Various features and aspects of the above-described examples may be used individually or jointly.

Further, while certain examples have been described using a particular combination of hardware and software, it should be recognized that other combinations of hardware and software are also possible. Certain examples may be implemented only in hardware, or only in software, or using combinations thereof. The various processes described herein may be implemented on the same processor or different processors in any combination.

Where devices, systems, components or modules are described as being configured to perform certain operations or functions, such configuration may be accomplished, for example, by designing electronic circuits to perform the operation, by programming programmable electronic circuits (such as microprocessors) to perform the operation such as by executing computer instructions or code, or processors or cores programmed to execute code or instructions stored on a non-transitory memory medium, or any combination thereof. Processes may communicate using a variety of techniques including but not limited to conventional techniques for inter-process communications, and different pairs of processes may use different techniques, or the same pair of processes may use different techniques at different times.

Specific details are given in this disclosure to provide a thorough understanding of the examples. However, examples may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the examples. This description provides example examples only, and is not intended to limit the scope, applicability, or configuration of other examples. Rather, the preceding description of the examples will provide those skilled in the art with an enabling description for implementing various examples. Various changes may be made in the function and arrangement of elements.

The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will, however, be evident that additions, subtractions, deletions, and other modifications and changes may be made thereunto without departing from the broader spirit and scope as set forth in the claims. Thus, although specific examples have been described, these are not intended to be limiting. Various modifications and equivalents are within the scope of the following claims.

In the foregoing specification, aspects of the disclosure are described with reference to specific examples thereof, but those skilled in the art will recognize that the disclosure is not limited thereto. Various features and aspects of the above-described disclosure may be used individually or jointly. Further, examples may be utilized in any number of environments and applications beyond those described herein without departing from the broader spirit and scope of the specification. The specification and drawings are, accordingly, to be regarded as illustrative rather than restrictive.

In the foregoing description, for the purposes of illustration, methods were described in a particular order. It should be appreciated that in alternate examples, the methods may be performed in a different order than that described. It should also be appreciated that the methods described above may be performed by hardware components or may be embodied in sequences of machine-executable instructions, which may be used to cause a machine, such as a general-purpose or special-purpose processor or logic circuits programmed with the instructions to perform the methods. These machine-executable instructions may be stored on one or more machine readable mediums, such as CD-ROMs or other type of optical disks, floppy diskettes, ROMs, RAMs, EPROMs, EEPROMs, magnetic or optical cards, flash memory, or other types of machine-readable mediums suitable for storing electronic instructions. Alternatively, the methods may be performed by a combination of hardware and software.

Where components are described as being configured to perform certain operations, such configuration may be accomplished, for example, by designing electronic circuits or other hardware to perform the operation, by programming programmable electronic circuits (e.g., microprocessors, or other suitable electronic circuits) to perform the operation, or any combination thereof.

While illustrative examples of the application have been described in detail herein, it is to be understood that the inventive concepts may be otherwise variously embodied and employed, and that the appended claims are intended to be construed to include such variations, except as limited by the prior art.

What is claimed is:

1. A computer-implemented method comprising:

accessing training data comprising a plurality of original utterances, wherein each original utterance of the plurality of original utterances comprises at least one named entity corresponding to a named entity category of a plurality of named entity categories;

accessing one or more gazetteers, wherein each gazetteer of the one or more gazetteers comprises a plurality of named entities extracted from the plurality of original utterances and a plurality of perturbed named entities derived from one or more named entities of the plurality of named entities;

generating a plurality of template utterances, wherein each template utterance of the plurality of template utterances comprises information from an original utterance of the plurality of original utterances and at least one placeholder identifier representing a named entity in the original utterance of the plurality of original utterances;

generating a plurality of additional utterances, wherein each additional utterance of the plurality of additional utterances comprises a template utterance of the plurality of template utterances populated with at least one named entity selected from a gazetteer of the one or more gazetteers;

augmenting the training data by adding the plurality of additional utterances to the plurality of original utterances; and

training a named entity recognition (NER) model with the augmented training data.

2. The computer-implemented method of claim 1, wherein each gazetteer of the one or more gazetteers corresponds to a different named entity category of the plurality of named entity categories.

3. The computer-implemented method of claim 1, wherein at least one gazetteer of the one or more gazetteers comprises a plurality of named entities retrieved from a source other than the training data.

4. The computer-implemented method of claim 3, wherein the plurality of named entities retrieved from the source other than the training data comprises named entities retrieved using at least one of a pre-trained model and a query-based search.

5. The computer-implemented method of claim 4, wherein the source other than the training data comprises at least one knowledge base.

6. The computer-implemented method of claim 1, wherein the plurality of perturbed named entities derived from one or more named entities of the plurality of named entities comprises perturbed versions of named entities of the plurality of named entities.

7. The computer-implemented method of claim 6, wherein a particular perturbed version of the perturbed versions of named entities of the plurality of named entities comprises a named entity having at least one typographical error.

8. The computer-implemented method of claim 1, further comprising:

providing the trained NER model to a system, wherein the providing the trained NER model includes detecting and classifying named entities in utterances received by the system from a user.

9. A system comprising:

one or more processors; and

one or more non-transitory computer-readable media storing instructions which, when executed by the one or more processors, cause the system to perform operations comprising:

accessing training data comprising a plurality of original utterances, wherein each original utterance of the plurality of original utterances comprises at least one named entity corresponding to a named entity category of a plurality of named entity categories;

accessing one or more gazetteers, wherein each gazetteer of the one or more gazetteers comprises a plurality of named entities extracted from the plurality of original utterances and a plurality of perturbed named entities derived from one or more named entities of the plurality of named entities;

generating a plurality of template utterances, wherein each template utterance of the plurality of template utterances comprises information from an original utterance of the plurality of original utterances and at least one placeholder identifier representing a named entity in the original utterance of the plurality of original utterances;

generating a plurality of additional utterances, wherein each additional utterance of the plurality of additional utterances comprises a template utterance of the plurality of template utterances populated with at least one named entity selected from a gazetteer of the one or more gazetteers;

augmenting the training data by adding the plurality of additional utterances to the plurality of original utterances; and

training a named entity recognition (NER) model with the augmented training data.

10. The system of claim 9, wherein each gazetteer of the one or more gazetteers corresponds to a different named entity category of the plurality of named entity categories.

11. The system of claim 9, wherein at least one gazetteer of the one or more gazetteers comprises a plurality of named entities retrieved from a particular source other than the training data.

12. The system of claim 11, wherein the plurality of named entities retrieved from the particular source other than the training data comprises named entities retrieved using at least one of a pre-trained model and a query-based search.

13. The system of claim 12, wherein the particular source other than the training data comprises at least one knowledge base.

14. The system of claim 9, wherein the plurality of perturbed named entities derived from one or more named entities of the plurality of named entities comprises perturbed versions of named entities of the plurality of named entities.

15. The system of claim 14, wherein a particular perturbed version of the perturbed versions of named entities of the plurality of named entities comprises a named entity having at least one typographical error.

16. The system of claim 9, the operations further comprising:

providing the trained NER model to a system, wherein providing the trained NER model includes detecting and classifying named entities in utterances received by the system in which the trained NER model is provided to from a user.

17. One or more non-transitory computer-readable media storing instructions which, when executed by one or more processors, cause a system to perform operations comprising:

accessing training data comprising a plurality of original utterances, wherein each original utterance of the plurality of original utterances comprises at least one named entity corresponding to a named entity category of a plurality of named entity categories;

accessing one or more gazetteers, wherein each gazetteer of the one or more gazetteers comprises a plurality of named entities extracted from the plurality of original utterances and a plurality of perturbed named entities derived from one or more named entities of the plurality of named entities;

generating a plurality of template utterances, wherein each template utterance of the plurality of template utterances comprises information from an original utterance of the plurality of original utterances and at least one placeholder identifier representing a named entity in the original utterance of the plurality of original utterances;

generating a plurality of additional utterances, wherein each additional utterance of the plurality of additional utterances comprises a template utterance of the plurality of template utterances populated with at least one named entity selected from a gazetteer of the one or more gazetteers;

augmenting the training data by adding the plurality of additional utterances to the plurality of original utterances; and

training a named entity recognition (NER) model with the augmented training data.

18. The one or more non-transitory computer-readable media of claim 17, wherein each gazetteer of the one or more gazetteers corresponds to a different named entity category of the plurality of named entity categories.

19. The one or more non-transitory computer-readable media of claim 17, wherein at least one gazetteer of the one or more gazetteers comprises a plurality of named entities retrieved from a particular source other than the training data.

20. The one or more non-transitory computer-readable media of claim 17, wherein the plurality of perturbed named entities derived from one or more named entities of the plurality of named entities comprises perturbed versions of named entities of the plurality of named entities.

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